Department of Sciences Faculty Members

Birla Institute of Technology, Mesra, Jaipur Campus

Dr. Anand Kumar Srivastava

Joined Institute on: 3-Nov-2009

Associate Professor and In-Charge

Physics

Ph.D., M.Tech., M.Sc.

CONTACT ADDRESS

27, Malviya Industrial Area, Jaipur, Rajasthan - 302017

Phone Office: 8107005029

Phone Residence: 8107005029

Email: aksrivastava@bitmesra.ac.in

PROFESSIONAL PROFILES

LinkedIn Profile: www.linkedin.com/in/dr-anand-kumar-srivastava-25529916

Google Scholar: https://scholar.google.com/citations?user=WedDCwwAAAAJ&hl=en&authuser=1

Web of Science: https://www.webofscience.com/wos/author/record/KTH-8218-2024

WORK EXPERIENCE

Teaching: 15 Years

Research: 22 Years

Individual: 22 Years

EDUCATIONAL QUALIFICATIONS

Doctor of Philosophy (Ph.D. in Physics)

Institute for Plasma Research (IPR), Gandhinagar. (Registered to Devi Ahilya Vishwavidhyalaya, Indore).

Thesis title: Experimental Studies on Atmospheric pressure glow discharge plasma

Thesis Supervisor: Dr. K. S. Ganesh Prasad

Master of Technology (M. Tech.) in Biomedical Engineering

IIT, BHU, Varansi, India.

Master of Science (M.Sc.) in Physics

PPN College, C.S.J.M. University, Kanpur, India.

INDUSTRIAL EXPERIENCE

Research & Development of Plasma Display Panel (PDP) at Samtel Color Limited, Ghaziabad (2006-2009).

Working Area at Industry:

- PDP cell structure and pixel design.
- Theoretical and equivalent circuit analysis of PDP performance.
- PDP gas discharge issues for luminous efficacy improvement.
- Electrostatic 2D field plotting and analysis using E-STAT simulator.
- Panel performance characterization.
- PDP manufacturing cost reduction.

SUBJECTS TAUGHT AT BIT MESRA, JAIPUR CAMPUS

- Physics Theory
- Physics Lab
- Electrical Measurement and Instrumentation
- Electronic Measurements
- Electromagnetic Theory
- Semiconductor Devices
- Understanding Harmony (UHV-II)
- Human Values and Professional Ethics
- Creative Arts

ACADEMIC/ADMINISTRATIVE RESPONSIBILITIES AT BIT JAIPUR CAMPUS

- In-charge of Department of Sciences (w.e.f. 01 September 2022).
- In-charge of Admissions in all UG/PG courses.
- Faculty in-charge of Training & Placements advisory committee.
- Member secretary of Career Advancement Scheme (CAS) Committee.
- Faculty Coordinator of AICTE Approval committee (November 2021 – April 2022).
- Member of Central Policy Committee (CPC).
- Member of Doctoral Committee for PhD evaluation.
- Member of Departmental Academic Committee (DAC).
- Member of SC/ST Cell.
- Member of Institute Innovation Council.
- Mentor of Science Club & Plasma Technology Club.
- In-charge of Plasma Lab.
- Physics subject coordinator at BIT Jaipur campus.
- Coordinator of National Science Day.
- Coordinator of National Education Day.

RESEARCH AREAS

Plasma Physics, Dielectric Barrier Discharge, Atmospheric Pressure Plasmas, Plasma Displays, Discharge Instabilities, Industrial Plasma Applications

ORGANIZATION OF CONFERENCE/SYMPOSIA/WORKSHOP

- Convener of 36th National Symposium on Plasma Science and Technology (PLASMA-2021), 13-15 December 2021 at BIT Mesra, Jaipur Off-Campus.
- Convener of IPR Outreach Program on Plasma, Its Applications and Nuclear Fusion, "PLASMA GYAAN-2022" (Training /Workshop for Students & Teachers on Basic Plasma Physics), 22-26 August 2022.
- Co-convener of Science section in International Conference on contemporary trends in multidisciplinary research and innovation (ICCTMRI – 2023) at BIT Mesra, Jaipur Off-Campus, 27-28 July 2023.
- Convener of International Workshop on Cold Plasma Technology and Applications, "CPTA-2025", at BIT Mesra, Jaipur Off-Campus, 6-8 February 2025.

WORKSHOPS/TRAINING/REFRESHER COURSE COMPLETED

- One week FDP on NEP 2020 Orientation and Sensitization Programme under Malviya Mission TTP of UGC (online), 27 May – 04 July 2025.
- One week FDP on Innovative Teaching & Learning Pedagogy organized by Research Foundation of India, 18-24 June 2025.
- One week FDP on Universal Human Values (UHV-II) organized by AICTE (Online), 27 May- 01 June, 2024.
- One week FDP on Universal Human Values (UHV-I) organized by AICTE (Online), 19-23 April, 2021.
- STEM Teacher Training Workshop on Research Based Pedagogical Tools, 23-26 September 2018, Kolkata (organized by IISER Pune).
- Joint IAEA-ICTP School on Atomic Processes in Plasmas during 27 Feb – 3 March 2017 at ICTP, Trieste, Italy.
- Science Academies 68th Refresher Course in Experimental Physics during 10-25 March 2015 (16 Days) at Central University of Rajasthan, Ajmer.
- DST-sponsored Faculty development Program on Entrepreneurship Development at BIT, Mesra, Jaipur Campus during 15-27 January 2013.
- General Management Program (GMP) at Samtel Color Limited, Ghaziabad.
- Presentation skills and Art of public speaking at Samtel Color Limited, Ghaziabad.

EXPERT TALK ON DD RAJASTHAN TV CHANNEL

A TV talk with me as Physics Expert entitled "Physics Education and Career Possibilities" has been broadcasted by Prasar Bharti, Doordarshan Kendra, Jaipur on 22.03.2014 in the 'Vigyan Jagat' program.

INVITED TALKS/ORAL PRESENTATIONS

1. The contributory oral talk entitled "Atmospheric pressure glow discharge for microwave invisibility: Experimental steps towards realization" delivered at PSSI National symposium on Plasma Science, December 2005, Cochin India.

2. Invited talk entitled "Aerodynamic applications of Atmospheric pressure glow discharge plasma" delivered at Department of Aerospace Engineering, Indian Institute of Science (IISc), Bangalore on 8th December 2005.

3. The Plasma Science Seminar entitled "Experimental study of atmospheric pressure glow discharge" presented at Plasma Science Laboratory, Department of Electrical and computer engineering, University of Tennessee, Knoxville, USA on 29th May 2006.

4. Invited talk entitled "Production and optimization of atmospheric pressure glow discharge for microwave attenuation" delivered at CZ Technologies, Lawrence, Kansas City, USA on 31st May 2006.

5. Invited talk entitled "Studies of Dielectric controlled Atmospheric pressure glow discharge for drag reduction and microwave dissipation" delivered at Department of Physics and Astronomy, University of Iowa, Iowa city, USA on 2nd June, 2006.

6. The contributory oral talk entitled "Observation of structures and striations in atmospheric pressure helium surface glow discharge" delivered at IEEE International Conference on Plasma Science, ICOPS-2006, Traverse city, Michigan, USA during 04-08 June, 2006.

7. Invited talk entitled "Microwave attenuation in uniform atmospheric pressure glow discharge plasma" delivered at Department of Physics, Central Michigan University, Mount Pleasant, USA on 9th June 2006.

8. Invited talk entitled "Industrial applications of Dielectric Barrier Discharges" delivered at Department of Microwave and Laser engineering, University of Michigan, Ann Arbor, Michigan, USA on 12th June 2006.

9. Nominated for an 'Invited talk' entitled "Non-thermal atmospheric pressure glow discharge plasmas for prospective applications" at Mahatma Gandhi University, Kottayam, Kerala in PPOMP-2011 on 27th November 2011.

10. The contributory oral talk entitled "Comparative Study of parallel-plate and planar-surface dielectric barrier glow discharge at atmospheric pressure" delivered at National Symposium on Plasma Science and Technology, PLASMA-2011, December 2011, BIT, Patna, India.

11. The contributory oral talk entitled "Dielectric barrier discharge plasma sources for Industrial research and development" delivered at National Conference on Vacuum Electronic Devices and Applications, VEDA-2012, 23rd September 2012, CSIR-CEERI, Pilani, India.

12. The contributory oral talk entitled "Evaluation of dielectric barrier discharge plasma sources for industrial applications" delivered at 103rd Indian Science Congress, University of Mysore, India, 5 January 2016.

13. Invited talk entitled "Patenting in plasma technology: example of plasma TV development in India" delivered at International conference on plasma science, technology and application (ICPSTA-2016), Amity University Lucknow, India, 21 January 2016.

14. Invited talk entitled "Atmospheric Pressure Non-thermal Plasmas for Biomedical Applications" delivered at National conference on Energy for sustainable development, Sagar University, India, 24 March 2017.

15. Invited talk entitled, "Selection of Dielectric Material for Producing Diffuse Dielectric Barrier Discharge Plasma at Atmospheric Pressure", delivered at International Conference on Nanotechnology: Idea, Innovations and Initiatives-2017 at IIT, Roorkee, 06 December 2017.

16. Invited talk entitled, "Dielectric Barrier Discharge Plasma Sources (Tutorial Lecture)" delivered at SERB School on Plasma Devices: Science and Technology at CSIR-CEERI, Pilani, Rajasthan, 12 December 2017.

17. Invited talk entitled, "Stationary and Moving Pattern formation in Surface Dielectric Barrier Discharge Plasma" delivered at 106th Indian Science Congress, LPU Jalandhar, 5th January 2019.

18. Invited talk entitled, "Atmospheric Pressure Cold Plasma Sources for Biomedical Applications" delivered at 12th International Conference on Plasma Science and Applications (ICPSA-2019), University of Lucknow, 13 November 2019.

19. The contributory oral talk entitled, "Atmospheric Pressure Glow Discharge Plasma Devices for Bio-Medical Applications" at International Conference on Energy & Environment: Challenges and Opportunities for Industries, MNIT Jaipur, 29 December 2019.

20. Invited talk entitled, "Prospects of Plasma Science & Technology" at St. Anselm's Pink City School, Jaipur, 30 April 2022.

21. The contributory online oral talk entitled, "Design Aspects of Atmospheric Pressure Cold Plasma Device for Societal Applications" at International Conference of Recent Trends in Engineering and Sciences (RTES-2023), 3 May 2023.

22. Invited talk on "Plasma Science & Technology" at Department of Physics, Rajasthan University, Jaipur, 05 December 2024.

RESEARCH AREAS

Plasma Physics, Dielectric Barrier Discharge, Atmospheric Pressure Plasmas, Plasma Displays, Discharge Instabilities, Industrial Plasma Applications

AWARD AND HONOURS

- Certificate of Reviewing -2020 from Journal of Electrostatics, Elsevier.
- Outstanding Reviewer Award -2018 from Journal of Physics D: Applied Physics, IOP Publishing, UK.
- ICTP Grant received to attend Joint IAEA-ICTP School on Atomic Processes in Plasmas, 27 February – 3 March 2017 by International Centre for Theoretical Physics (ICTP), Trieste, Italy.
- Bharat Shiksha Ratan Award-2014 for academic excellence conferred by GSHEG, Delhi during a National Seminar on 10th Feb 2014.
- ISCA Best Paper (Poster) Presentation Award in Physical Sciences during 101st Indian Science Congress, University of Jammu, 3-7 February 2014.
- DST, Govt of India-International Travel support (ITS) for young scientists, received to attend 4th International Conference on Plasma Medicine (ICPM-4) at Orleans, France, 17-21 June 2012.
- PSSI-Best Paper (poster) Award-2010 in Plasma Physics during PSSI national symposium, PLASMA-2010, Guwahati.
- CSIR-UGC (NET) JRF-June 2002, Examination (Physics) qualified.

- JEST-2001 Examination (Physics) qualified.
- BHU Medal for standing first at M.Tech. Examination in Biomedical Engineering 2001.
- GATE-1999 Examination (Physics) qualified.
- Government scholarships at college level.

PUBLICATIONS

JOURNAL PAPERS:

1. A. K. Srivastava, G. Prasad, Vinay Kumar, M. B. Chowdhari and Ram Prakash, "Experimental studies on Atmospheric pressure glow discharge for drag reduction and microwave invisibility" Journal of Aerospace Sciences and Technologies, Vol. 57, No. 1, 17-23, Feb (2005).

2. A. K. Srivastava, M. K. Garg, G. Prasad, Vinay Kumar, M. B. Chowdhari and Ram Prakash, "Characterization of atmospheric pressure glow discharge in helium using Langmuir probe, emission spectroscopy and discharge resistivity" IEEE Transaction on Plasma Sciences, Vol. 35, Issue 4, 1135(2007).

3. A. K. Srivastava, G. Prasad, P. K. Atrey and Vinay Kumar, "Attenuation of microwaves propagating through parallel-plate helium glow discharge at atmospheric pressure", Journal of Applied Physics, Vol. 103, 033302 (2008).

4. A. K. Srivastava and G. Prasad, "Spatio-temporal dynamics of plasma spots in helium surface barrier discharge", Physics Letters A, Vol. 372, 6101-6106 (2008).

5. A. K. Srivastava and G. Prasad, " Low frequency ionization waves in helium surface barrier glow discharge at atmospheric pressure", Physics Letters A, Vol. 374, 3960-3966 (2010).

6. Maharshi Samanta, A. K. Srivastava, S. Sharma, A. Rastogi and H. K. Dwivedi, "Electrical discharge as an inspection method for imperfect plasma display cells", IOP Journal of Physics: Conference Series 208 (2010) 012113.

7. Shashank Sharma, A. K. Srivastava, Harmeet Singh, M. Raja and H. K. Dwivedi, "Optimization of sustain pulse parameters for luminous efficacy improvement in an AC plasma display panel", Displays, Elsevier 31, 122-127 (2010).

8. A. K. Srivastava, Shashank Sharma and H. K. Dwivedi, "Electrostatic analysis for designing of efficient discharge cell structures in AC plasma displays", Journal of Electrostatics, 68, 394-399 (2010).

9. A. K. Srivastava and G. Prasad, "Characteristics of uniform parallel-plate and planar-surface dielectric barrier glow discharge at atmospheric pressure", Journal of Electrostatics, Vol.72, 140-146 (2014).

10. Anand Kumar Srivastava, "Observation of self-organized square lattice pattern formation in surface barrier discharge plasma", Journal of Electrostatics, Vol.82, 72-78 (2016).

11. Anand Kumar Srivastava, "Dielectric Erosion and Sub-micron Polymeric Particles Formation in Planar Dielectric Barrier Discharge Plasma", Advanced Science Letters, Vol.22, No.11, 3879-3881 (2016).

12. Ravindra Kumar, A. K. Srivastava and S. Konar, "Evaluation of surface properties of various plasma nitride low alloy steels", Advanced Science Letters, Vol.22, No.11, 3919-3923 (2016).

13. Anand Kumar Srivastava, "Selection of Dielectric Material for Producing Diffuse Dielectric Barrier Discharge Plasma at Atmospheric Pressure", Materials Today: Proceedings, Elsevier, Vol 18, 1033-1038, 2019.

14. S. Patel, A. K. Srivastava, M.B. Chowdhuri, R. Manchanda, J. Ghosh, A. Bhattacharya, J. Raval, U. Nagora, P.K. Atrey and R. L. Tanna, "Study of Iron impurity behaviour using VUV spectroscopy in ADITYA and ADITYA-U Tokamak" NUCLEAR FUSION, 59, 086019, 2019.

15. M. B. Chowdhuri, Anand K. Srivastava et. al, "Impurity Behaviour in high performance ADITYA tokamak plasmas", PLASMA AND FUSION RESEARCH, 17, 2402011, 2022.

16. Anand Kumar Srivastava, "Microwave Attenuation Measurement as a Diagnostic Method to Estimate Electron Density in Planar Surface Barrier Discharge Plasma", Indian Journal of Physics, Springer, https:/doi.org/10.1007/s12648-023-02988-2, 2023.

CONFERENCE PAPERS (FULL-LENGTH):

1. "Experimental studies on Atmospheric pressure glow discharge for drag reduction and microwave invisibility" in proceedings of AERO-INDIA 2005, International Seminar, Bangalore, India, February 2005.

2. "Atmospheric pressure glow discharge for microwave invisibility: Experimental steps towards realization" in proceedings of the National Symposium on Plasma science and Technology, PLASMA-2005, Cochin, India, December 2005.

3. "Development of low-cost ITO-less fence electrode PDP cell structure" PDP 1-3, 'Cell Technology' in proceedings of International Display Workshop, IDW-2008, at Niigata, Japan, 05-08 December 2008.

4. "Progress on Development of Low Temperature Processing of Low-Cost ITO-free PDP using Soda-lime-silica Glass Substrate" PDP1-2, 'Novel PDPs' in the proceedings of International Display Workshop, IDW-2009, at Miyazaki, Japan, 8-10 January, 2009.

5. "Effect of vacuum level and gaseous impurities on performance of Plasma Display TV" in proceedings of National Conference on Vacuum Electronic Devices and Applications, VEDA-09, at IT-BHU, Varansi, January, 2009.

6. "Design study of discharge cell structure for efficiency improvement in AC plasma display TV" in proceedings of National Conference on Vacuum Electronics Devices and Applications, VEDA-09, at IT-BHU, Varansi, January, 2009.

7. "Fast Rise-time asymmetric bipolar waveform for driving high luminous efficacy plasma displays" in the proceedings of Asia Display, IMID-2010, at Kintex, Seoul, Korea, 11-15 October, 2010.

8. "Non-thermal atmospheric pressure glow discharge plasmas for prospective applications", in the proceedings of "First International Conference on Plasma Processing of Organic Materials and Polymers, PPOMP-2011" at Kottayam, Kerala, India, 25-27 November 2011.

9. "Dielectric barrier discharge plasma sources for Industrial research and development" in the proceedings of National Conference on Vacuum Electronic Devices and Applications, VEDA-2012, at CSIR-CEERI, Pilani, Rajasthan, September, 2012.

10. "Optimization study of dielectric barrier discharge plasma cells for efficient UV/VUV emission" in the proceedings of National Symposium on Plasma Science and Technology, PLASMA-2012, Pondicherry, India, December 2012.

11. "Patenting in plasma technology: example of plasma TV development in India" in the proceedings of International conference on plasma science, technology and application (ICPSTA-2016), Lucknow, India, January 2016.

12. "Development of dielectric-controlled cold atmospheric pressure plasma sources" in the proceedings of International Conference on Engineering Physics, Materials and Ultrasonics (ICEPMU), NCU Gurgaon, India, June 2016.

13. S. Patel, A. K. Srivastava, M.B. Chowdhuri, R. Manchanda, J. Ghosh, A. Bhattacharya, J. Raval, U. Nagora, P.K. Atrey And R. L. Tanna, "Study of Iron impurity behaviour using VUV spectroscopy in ADITYA and ADITYA-U Tokamak" (EX-P4-3 (1-7)) in the Proceedings of 27th IAEA Fusion Energy Conference, Ahmedabad, India, 22-27 Oct, 2018.

14. Anand Kumar Srivastava, "Design Aspects of Atmospheric Pressure Cold Plasma Device for Societal Applications" in the proceedings of International Conference on Recent Trends in Engineering and Sciences (RTES – 2023), NIT Surat, 2-3 May, 2023.

CONFERENCE PAPERS: ABSTRACTS (POSTER PRESENTATIONS):

1. "Experimental study of atmospheric pressure glow discharge plasma" in proceedings of the National Symposium on Plasma Science and Technology, PLASMA-2002, Coimbatore, India, December 2002.

2. "Production of surface glow discharge at atmospheric pressure and observation of waves with the formation of plasma bubbles" in proceedings of the National Symposium on Plasma Science and Technology, PLASMA-2003, Ranchi, India, December 2003.

3. "Comparative study of plasma diagnostics in helium atmospheric pressure glow discharge using Langmuir probe and emission spectroscopy" in proceedings of International congress on Plasma Physics, ICPP-2004, Nice, France, 2004.

4. "Observation of Acoustic wave phenomenon in Atmospheric pressure surface glow discharge" in proceedings of International conference on Frontiers of Plasma Physics and Technology, FPPT-2005, Goa, India, 2005.

5. "Experimental study of microwave interaction with collisional glow discharge plasma" in proceedings of National Symposium on Plasma Science and Technology, PLASMA-2005, Cochin, India, December 2005.

6. "Observation of localized plasma structures in planar surface barrier discharge" in proceedings of the National Symposium on Plasma Science and Technology, PLASMA-2005, Cochin, India, December 2005.

7. "Observation of structures and striations in atmospheric pressure helium surface glow discharge" in proceedings of IEEE International Conference on Plasma Science, 1D5, ICOPS-2006, Traverse city, Michigan, USA, 04-08 June, 2006.

8. "Dissipation of microwaves propagating through atmospheric pressure glow discharge plasma" in proceedings of IEEE International Conference on Plasma Science, 3P70, ICOPS-2006 at Traverse city, Michigan, USA, 04-08 June, 2006.

9. "Microwave attenuation measurement as a diagnostic tool for atmospheric pressure glow discharge plasma" in proceedings of the national symposium on plasma sciences, PLASMA-2006, MNIT, Jaipur, India, December 2006.

10. "Observation of self-excited ionization waves in a planar dielectric barrier discharge system at atmospheric pressure" in proceedings of the National Symposium on Plasma Science and Technology, PLASMA-2010, Guwahati, India, December 2010.

11. "Comparative Study of parallel-plate and planar-surface dielectric barrier glow discharge at atmospheric pressure" in proceedings of the National Symposium on Plasma Science and Technology, PLASMA-2011, BIT, Patna, India, December 2011.

12. "DBD Cell Designs for Efficient UV/VUV Production Suitable to Biomedical Treatment and Sterilization," in the proceedings of 4th International Conference on Plasma Medicine, ICPM-4, Orleans, France, June 2012.

13. "Design and Characteristics of atmospheric-pressure dielectric barrier discharge plasma sources for daily-life requirements" in the proceedings of 101st Indian Science Congress, Jammu, India, February 2014.

14. "Observation of Self-organized Square Lattice Pattern in Planar Surface Barrier Discharge Plasma at Atmospheric Pressure" in the proceedings of 102nd Indian Science Congress, Mumbai, India, January 2015.

15. "Evaluation of dielectric barrier discharge plasma sources for industrial applications" in the proceedings of 103rd Indian Science Congress, Mysore, India, January 2016.

16. "Atmospheric Pressure Non-thermal Plasmas for Biomedical Applications" in the proceedings of National conference on Energy for sustainable development, Sagar, India, March 2017.

17. "Estimation of Emissivity of Fe14+ and Fe15+ VUV Spectral Lines Relevent to Aditya-U Tokamak Plasma" (343/PD-06/P) in the proceedings of 32nd National Symposium on Plasma Science and Technology (PLASMA-2017), IPR, Gandhinagar, Gujarat, 07-10 November 2017.

18. "Simulation of Radial Profile Of Fe14+ &Fe15+ Emissions In Aditya-U Tokamak Plasma" (PD-06) in the proceedings of NFP-PFRC Vision Meeting, Nirma University, Ahmedabad, Gujarat, 26-28 April, 2018.

19. "Modeling of Experimental VUV Spectra from ADITYA-U Tokamak" in proceedings of 33rd National Symposium on Plasma Science and Technology (PLASMA-2018), University of Delhi, 4-7 December 2018.

20. "Stationary and Moving Pattern formation in Surface Dielectric Barrier Discharge Plasma" in the proceedings of 106th Indian Science Congress, Lovely Professional University, Punjab, India, 3-7 January 2019.

21. "Atmospheric Pressure Cold Plasma Sources for Biomedical Applications" in the proceedings of 12th International Conference on Plasma Science and Applications (ICPSA-2019), University of Lucknow, 11-14 November 2019.

22. "Atmospheric Pressure Glow Discharge Plasma Devices for Bio-Medical Applications" in the proceedings of International Conference on Energy & Environment: Challenges and Opportunities for Industries, MNIT Jaipur, 28-29 December 2019.

23. "Estimation of Iron Impurity Transport Parameters in Aditya Tokamak Plasma" in the proceedings of 107th Indian Science Congress, University of Agricultural Sciences, Bangalore, India, 3-7 January 2020.

24. "Iron Impurity Behavior Study in the ADITYA Tokamak" in the proceedings of National Symposium for Commemorating 30-years of Aditya Tokamak at IPR Gandhinagar, 27-28 January 2020.

25. "Utilization of Dilelctric Barrier Discharge Plasma for Benefit of Mankind" in the proceedings of 36th National Symposium on Plasma Science and Technology (PLASMA-2021), BIT Mesra, Jaipur Campus, 13-15 December, 2021.

26. "Breakdown Characteristics of a Non-Thermal Low Pressure Glow Discharge Plasma" will come in the proceedings of International Conference on contemporary trends in multidisciplinary research and innovation (ICCTMRI – 2023) at BIT Mesra, Jaipur Off-Campus, 27-28 July 2023.

27. "Evaluation of Glow Discharge Plasma Properties produced at atmospheric pressure" will come in the proceedings of International Conference on contemporary trends in multidisciplinary research and innovation (ICCTMRI – 2023) at BIT Mesra, Jaipur Off-Campus, 27-28 July 2023.

MEMBER OF PROFESSIONAL BODIES

- Life member of PSSI, Plasma Science Society of India.
- Life member of Association of Asia Pacific Physical Societies, Division of Plasma Physics (AAPPS-DPP), Japan.
- Member of ISPM, International Society of Plasma Medicine, USA.
- Former student member (2006-2007) of IEEE society, USA.
- Former student member (2006-2007) of IEEE communication society, USA.
- Life member of Indian Science Congress Association (ISCA).
- Member of Global Society for Health and Educational Growth, New Delhi.
- Life member of Indian Association of Physics Teachers (IAPT).
- Life member of Indian Physics Association (IPA).

CURRENT SPONSORED PROJECTS

Project title: Studies of High Z impurity behavior using VUV spectral emissions from the tokamaks in IPR.

Duration: 2017-2020

Funding Agency: BRNS, Department of Atomic Energy, Govt. of India

Sanctioned Grant: 25.6 Lakh

Status: Completed

Role: Principal Investigator

INTELLECTUAL PROPERTY RIGHTS (IPR)

Patent Applications filed:

1. The patent entitled "A plasma display panel and its manufacturing thereof" filed at Indian patent office, New Delhi on August 6, 2007 with application no. 1885/DEL/2007.

2. The patent entitled "A plasma display panel and with reduced address voltage" filed at Indian patent office, New Delhi on October 25, 2007 with application no. 2341/DEL/2007.

3. The patent entitled "A plasma display panel and enhanced luminance efficacy" filed at Indian patent office, New Delhi on October 19, 2007 with application no. 2188/DEL/2007.

4. The patent entitled "A plasma display panel and its manufacturing thereof" filed at Indian patent office, New Delhi on March 10, 2008 with application no. 598/DEL/2008.

5. The patent entitled "Plasma display panel and method for manufacturing the same" filed at Indian patent office, New Delhi on December 02, 2008 with application no. 2719/DEL/2008.

6. The patent entitled "Plasma display apparatus and method for driving thereof" filed at Indian patent office, New Delhi on July 16, 2009 with application no. 1459/DEL/2009.

MEMBER, EDITORIAL BOARD

Reviewed Papers for the following international journals:

1. Journal of Applied Physics
2. Applied Physics Letters
3. Plasma Sources Science and Technology
4. Journal of Physics D: Applied Physics
5. Journal of Electrostatics, Elsevier

Dr. Gurminder Singh Malhotra

Joined Institute on: 1-Jul-2004

Associate Professor

Mathematics

Ph.D (2009) (University of Rajasthan), M.Sc (Mathematics, University of Rajasthan)

CONTACT ADDRESS

27, Malviya Industrial Area, Jaipur-302017

Phone Office: 9828854498

Phone Residence: 9828854498

Email: gurmindersingh@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 20 Years

Research: 10 Years

PROFESSIONAL BACKGROUND

SUBJECTS TAUGHT

- Mathematics 1 and Mathematics 2 (MA 24101 and MA24103)
- Probability and Statistics (AI201)
- Discrete Mathematics (MA205)
- Probability and Random Process (EC213)
- Mathematics for Data Science (AI203)
- Mathematics for Computing 1 and II (CN131)
- Numerical Methods (MA 204)

RESEARCH AREAS

Heat and Mass Transfer in Boundary Layer Flows

PROFESSIONAL PROFILES

Google Scholar: https://scholar.google.com/citations?hl=en&user=VP5CMagAAAAJ

LinkedIn: https://www.linkedin.com/in/gurminder-singh-68741a2a/

PUBLICATIONS

1. U.Mishra and Gurminder Singh, "Dual solution of forced convection along a stretching sheet with variable thickness in presence of free stream and magnetic field", Sians Malaysiana, Vol. 46(2), pp. 349-358, 2017.

2. U.Mishra and Gurminder Singh, "A study of mixed convection flow over stretching cylinder in presence of slip flow and thermal jump boundary conditions". International Journal of Fluid Mechanics Research, Vol. 43(4), pp.308-318, 2016.

3. U.Mishra and Gurminder Singh, "Convection along Unsteady shrinking sheet with boundary slip and thermal jump in presence of radiation", Acta Technica, Vol. 60(4), pp.405-416, 2016.

4. Gurminder Singh and O.D. Makinde "Mixed convection slip flow with temperature jump along a moving plate in the presence of free stream", Thermal Science., Vol. 19(1), pp.119-128, 2015.

5. U. Mishra and Gurminder Singh "MHD mixed convection and mass transfer due to unsteady stretching sheet", Heat Transfer-Asian Research (Wiley), Vol.43(5), pp. 447-458, 2014.

6. Gurminder Singh and O.D. Makinde, "Axisymmetric slip flow on a vertical cylinder with heat transfer". Sians Malaysiana, Vol. 43(3), pp.483-489, 2014.

7. U. Mishra and Gurminder Singh "Dual solutions of mixed convection flow with momentum and thermal slip flow over a permeable shrinking cylinder", Computer and Fluids, Vol. 93, pp. 107-115, 2014.

8. Gurminder Singh and P.R. Sharma "Heat and mass transfer in the boundary layer flow along a vertical isothermal reactive plate near stagnation point: existence of dual solution", Journal of Applied Fluid Mechanics, Vol. 7 (1), pp. 25-33, 2014.

9. Gurminder Singh and A.J. Chamkha "Dual solutions for second-order slip flow and heat transfer on a vertical permeable shrinking sheet", Ain Shams Engineering Journal, Vol.4, pp.911-917, 2013.

10. Gurminder Singh and O.D. Makinde "MHD slip flow of viscous fluid over an isothermal reactive stretching sheet", Annals of Faculty Engineering Hunedoara- International Journal of Engineering, Tome XI, Fascicule 2, pp. 41-45, 2013.

11. U. Mishra and Gurminder Singh "A similarity solution of a mixed convection flow along an unsteady stretching sheet", Caspian Journal of Applied Sciences Research, Vol. 2(3), pp. 17-23, 2013.

12. P.R. Sharma, Gurminder Singh and A.J. Chamkha "Steady mixed convection flow of water at 4°C along a non-isothermal vertical moving plate with transverse magnetic field" International Journal of Industrial Mathematics, Vol. 4(3), pp. 171-186, 2012.(ISSN 2008-5621)

13. Gurminder Singh, P.R. Sharama and A.J. Chamkha "Mass transfer with chemical reaction in MHD mixed convective flow along a vertical stretching sheet", International Journal of Energy and Technology, Vol. 4, Article 20(1-12), 2012

14. Gurminder Singh and Makinde, O.D. "Computational dynamics of MHD free convection flow along an inclined plate with Newtonian heating in the presence of volumetric heat generation". Chemical Engineering Communications, Vol. 199(9), pp.1144-1154, 2012. (ISSN 0098-6445)

15. Sharma, P.R., Gurminder Singh and Chamkha, A.J. "Unsteady heat transfer in steady stagnation point flow along a stretching sheet in the presence of free stream". International Journal of Heat and Technology, Vol.29 (1), pp.39-44, 2011.(ISSN 0392-8764)

16. Sharma, P.R. and Gurminder Singh "Steady MHD natural convection flow with variable electrical conductivity and heat generation along an isothermal vertical ". Tamkang Journal of science and Engineering, Vol. 13 (3), pp. 235-242, 2010. (ISSN 1560-6686)

17. Gurminder Singh, Sharma, P.R. and Chamkha, A.J. "Effect of volumetric rate of heat generation/ absorption on mixed convection stagnation point flow on an iso-thermal vertical plate in Porous media". International Journal of Industrial Mathematics, Vol. 2(2), pp. 59-71, 2010.(ISSN 2008-5621)

18. Sharma, P.R and Gurminder Singh. "Effects of Variable Thermal Conductivity, Viscous Dissipation on Steady MHD Natural Convection Flow of Low Prandtl Fluid on an Inclined Porous Plate with Ohmic Heating". Meccanica (Springer). Vol.45 (2), pp. 237-247, 2010.(ISSN 0025-6455)

19. Gurminder Singh, Sharma, P.R. and Chamkha, A.J. "Effect of thermally stratified ambient fluid on MHD convective flow along a moving non-isothermal vertical plate". International Journal of Physical Sciences, Vol. 5(3), pp. 208-215, 2010. (ISSN 1992-1950)

20. Sharma, P.R and Gurminder Singh. "Effects Of Varying Viscosity And Thermal Conductivity On Steady MHD Free Convective Flow And Heat Transfer Along An Isothermal Vertical Plate With Internal Heat Generation", International Journal for Numerical Methods for Heat and Fluid Flow, Vol. 19(1), pp. 78-92, 2009. (ISSN 0961-5539)

21. Sharma, P.R and Gurminder Singh. "Effects of Ohmic Heating and Viscous Dissipation on Steady MHD Flow Near a Stagnation Point on an Isothermal Stretching Sheet". Thermal Science International Scientific Journal, Vol. 13(1), pp.5-12, 2009. (ISSN 0354-9836)

22. Sharma, P.R and Gurminder Singh. "Effect of Variable Thermal Conductivity and Heat Source / Sink on MHD Flow Near a Stagnation Point on a Linearly Stretching Sheet". Journal of Applied Fluid Mechanics, Vol. 2, No. 1, pp. 13-21, 2009. (ISSN 1735-3645)

23. Sharma, P.R and Gurminder Singh. "Numerical Solution of Transient MHD Free Convection Flow of an Incompressible Viscous Fluid Along an Inclined Plate With Ohmic Dissipation". International Journal of Applied Mathematics and Mechanics. Vol.5(5), pp. 57 - 65, 2009. (ISSN 0973-0184)

24. Sharma, P.R and Gurminder Singh. "Unsteady Flow About A Stagnation Point On A Stretching Sheet In The Presence Of Variable Free Stream", TIJSAT, Vol. 13(1), pp. 11-16, 2008. (ISSN 0859-4074).

25. Sharma, P.R and Gurminder Singh. "Unsteady MHD Free Convective Flow and Heat Transfer Along a Vertical Porous Plate with Variable Suction and Internal Heat Generation". International Journal of Applied Mathematics and Mechanics, Vol. 4(5), pp. 1-8, 2008.(ISSN 0973-0184)

26. Sharma, P.R and Gurminder Singh. "Effects of Viscous Dissipation and Heat Source / Sink on Steady MHD Flow Near a Stagnation Point on a Linearly Stretching Sheet Subjected to Different Thermal Boundary Conditions". A.M.S.E. Journal (Modelling B), Vol. 77(1), pp. 35-53, 2008.(ISSN 1259-5969)

PAPERS IN CONFERENCES

1. Gurminder Singh and Peeyush Tewari, "Modelling the total income of a Hindi movie using Modified Exponential Model", Paper T 303, International conference on the recent Trends in Engineering and Sciences (2-3 May, 2023), Organized by N.I.T, Surat.

2. Gurminder Singh, "Mass transfer in laminar boundary layer flow over a vertical flat surface with convective boundary condition in presence of uniform free stream", at National Conference on Computational and Mathematical Science, VIT Campus, Jaipur.

3. Gurminder Singh, "A similarity solution of laminar boundary layer flow over a vertical flat surface with convective boundary condition" at National Conference "Numerical & Mathematical Modeling in Electrical Engineering & Applied Sciences - 2012 " at Poornima College of Engineering.

4. Gurminder Singh, "Numerical solution for mixed convection flow along a moving vertical plate with Newtonian surface heating", at UGC Sponsored- National Seminar on "Recent trends in Fluid dynamics and Optimization Techniques and 15th Prof. P.D. Verma Memorial Lecture", University of Rajasthan, Jaipur.

5. Gurminder Singh "Natural Convection along an inclined plate with Newtonian heating in the presence of transverse magnetic field" at 13th International Conference of International Academy of Physical Science (CONIAPS XIII) organised by University of Petroleum and Energy Studies, Dehradun.

6. Gurminder Singh "Heat transfer characteristics of a continuously stretching sheet with the sudden change in surface temperature" at National Symposium on "Application of various techniques in fluid dynamics"(UGC Sponsored) organized by Department of Mathematics, B.S.N.V PG College, Lucknow.

7. Gurminder Singh, P.R. Sharma and A.J.Chamkha, "Stagnation point mixed convection flow and mass transfer in porous medium along a non-isothermal permeable vertical plate in presence of heat source/sink and chemical reacting specie", at International Congress of Mathematicians 2010 at University of Hyderabad, Hyderabad.

8. Gurminder Singh "Numerical Solution of laminar flow and heat transfer between two parallel horizontal sheets", at National Seminar on Recent Development in Fluid Dynamics and 14th P.D. Verma Memorial Lecture, University of Rajasthan, 2010.

9. Gurminder Singh, Vivek Bhola and Mahesh Grover "Detecting Communities of Practice in Organisational Network: A Bibliometric Approach" at Guru Jambheshwar University of Science and Technology, Hisar, 2010.

10. Gurminder Singh "Variable Viscosity Effect on MHD Mixed Convection Flow Along a Continuously Moving Vertical Plate With Variable Surface Temperature" at Guru Jambheshwar University of Science and Technology, Hisar, 2010.

11. Gurminder Singh "Numerical Solution of Unsteady Boundary Layer Flow And Heat Transfer on a non-Isothermal Stretching Sheet", at B.K.B.I.E.T, Pilani, 2009.

12. Sharma,P.R and Gurminder Singh. "A Computational Scheme for Free Convective Flow and Heat Transfer Through a Viscous Incompressible Fluid Along a Vertical Plate", "National Conference on Biomechanics" at IIT, Roorkee, 2009.

INVITED TALK/LECTURE

1. Delivered a Talk on "Mathematical Foundations of Generative AI" at ATAL Faculty Development Program on "Generative Artificial Intelligence : Tools and Techniques" conducted at B.I.T Jaipur Campus from 25th November to 30th November, 2024.

2. Presented a talk on "A Review of numerical scheme for solving coupled non-linear differential equations arising in boundary layer flow" at S.V.N.I.T Surat in One week short term training Program.

3. Dual Solutions of mixed convection flow with slip over a permeable shrinking cylinder: Understanding the Numerical scheme involved, at National conference on "Advances of Mathematics-2014" from Nov. 21-23, 2014 at Amity University.

WORKSHOP/TRAINING COURSES CONDUCTED/ORGANISED

1. Conducted Workshop on "Markov Chain and Hidden Markov Model" at B.I.T, Jaipur Campus, from 2nd December to 6th December, 2024; for undergraduate Engineering students. The workshop offered understanding a mathematical way for word suggestion in Generative AI.

2. Conducted Winter School on "Linear Algebra and Applications- Part I" for B.I.T. Jaipur Campus Undergraduate Students from 17 to 21, December 2103. The course taught "Matrices and Gaussian Elimination, Solving system of Linear Equations, Vector Spaces, Basis, Dimension, Projection of vector in a space, Orthogonality of Vectors and Least Square Approximation Method".

WORKSHOP/TRAINING/CONFERENCE ATTENDED

1. Two week "Refresher course in Mathematics" organized by Department of Mathematics, Ramanujan College, University of Delhi and Ministry of Education, from 31 August to 14 September, 2021.

2. One week "National Instructional Workshop in Cryptology 2015" by Cryptology Research Society of India at Himachal University, Shimla from May 18 to 22, 2015.

3. Two week ISTE main Workshop on "Fluid Mechanics" from 20-30 May, 2014 conducted by IIT Kharagpur at Remote center in Jaipur.

4. NBHM Sponsored conference on "Harmonic Analysis and Operator Theory" from 21-23 March, 2014 at IISER Bhopal.

5. One-week short term training Program on "MATLAB Programming for engineers" from Dec. 11 to Dec. 15, 2013 at N.I.T, Kurukshetra.

6. One-week short term training Program on "Significant role of Mathematical analysis in Applied Science and Engineering" from Sep. 30 to Oct 4, 2013 at S.V.N.I.T Surat.

7. Two week ISTE main workshop on "Computational Fluid Dynamics", from 12-22 June, 2012, conducted by I.I.T, Bombay at Remote center in Jaipur.

8. National Conference on Computational and Mathematical Sciences, COMPUTATIA-2012, from Jan 28-29, 2012, at V.I.T. Campus, Jaipur.

9. National Conference on "Numerical & Mathematical Modeling in Electrical Engineering & Applied Sciences - 2012 " at Poornima College of Engineering on Jan. 21, 2012.

10. UGC Sponsored- National Seminar on "Recent trends in Fluid dynamics and Optimization Techniques and 15th Prof. P.D. Verma Memorial Lecture" from Aug 6 to 7, 2011 at University of Rajasthan, Jaipur.

11. 13th conference of International Academy of Physical Sciences on "Emerging Interfaces of Physical Sciences" from 14-16 June, 2011 organized by University of Petroleum and Energy Studies, Dehradun.

12. National workshop on "Challenges before applied mathematician: Fluid Dynamics and Optimization Techniques" (UGC sponsored) from March 11 to 13, 2011 at University of Rajasthan.

13. National Symposium on "Application of various techniques in fluid dynamics"(UGC Sponsored) from Feb. 10 to 12, 2011, organized by Department of Mathematics, B.S.N.V PG college, Lucknow in association with National Science Network.

14. 12th conference of International Academy of Physical Sciences on "Emerging Interfaces of Physical Sciences" from 22-24 Dec., 2010 organized by University of Rajasthan, Jaipur.

15. "International Congress of Mathematicians, 2010", from August 19 to 27, 2010 organized by IMU & University of Hyderabad, Hyderabad.

16. "National Seminar on Recent Development in Fluid Dynamics and 14th P.D. Verma Memorial Lecture" from 7-8, Aug. 2010 at University of Rajasthan, 2010.

17. "National Conference on Mathematical Modelling and Simulation" from March 20 to 21, 2010 at Guru Jambheshwar University of Science and Technology, Hisar.

18. "National Seminar on Computational Mathematics and Operation Research", from November 28 to 29, 2009 at B.K.B.I.E.T, Pilani.

19. "National Conference on Biomechanics", from March 7-8, 2009, at IIT, Roorkee.

20. Indo-Australian Workshop on "CFD Approach on Fluid Flow, Heat and Mass Transfer" from April 12-14, 2007 at IIT, Roorkee.

21. Short Term Course on "Mathematical Methods in Engineering and Science", from July 3 to July 15, 2006 at IIT, Kanpur conducted by Centre for Development of Technical Education.

22. "Faculty Enablement Workshop on Soft Skills" at Birla Institute of Technology, Mesra from Dec 21-24, 2006.

23. Workshop on "Scientific computation, Numerical Analysis and Applications" from Jan.24 to Feb 04, 2005 at Indian Institute of Sciences, Banglore.

MEMBERSHIP

- Life Member of "Indian Society of Theoretical and Applied Mechanics", IIT, Kaharagpur.(L/529).
- Life Member of "Rajasthan Ganith Parishad", Ajmer.
- Life member of "National Society of Biomechanics"

TRAINING/COURSES

- 6 month certificate course in Software Technologies by Sky Web India Ltd in 2001.

Dr. (Mrs.) Bhawna Agarwal

Joined Institute on: 21-Mar-2009

Assistant Professor

Applied Sciences

Ph.D.

CONTACT ADDRESS

164/134 pratap nagar, sanganer, sector 16, jaipur

Phone Office: 01414019816

Phone Residence: +919782135064

Email: bhawnaagarwal@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 25 Years

Research: 5 Years

PROFESSIONAL BACKGROUND

EDUCATIONAL QUALIFICATIONS

Ph.D (Mathematics)

University of Rajasthan, Jaipur, India. May, 2008

M.A (Mathematics)

University of Delhi, India, 2000

B.A. Honours (Mathematics)

University of Delhi, India, 1997

Diploma in Business Management

I.M.T. Ghaziabad, India, July 2007.

SUBJECTS OF SP 2025

1. MATHEMATICS-2
2. NUMERICAL METHOD
3. NUMERICAL METHOD LAB

PROFESSIONAL PROFILES

LinkedIn: https://www.linkedin.com/in/bhawna-agarwal-420424272/

Google Scholar: https://scholar.google.com/citations?view_op=new_profile&hl=en&authuser=1

RESEARCH AREAS

Title: Laminar Boundary Layer flow problems of fluid dynamics with special reference to heat transfer.

PUBLICATIONS

REFEREED NATIONAL AND INTERNATIONAL JOURNAL PAPERS

1. P.R.Sharma and Bhawna Agarwal "Effects of heat sources on Unsteady bounded layer flow and heat transfer along a porous flat plate in the presence of variable suction and uniform free stream," J. Nat. Acad. Math., Vol.19, 2005, pp.-41-66.

2. P.R.Sharma and Bhawna Agarwal "Pulsatile MHD flow and heat transfer between parallel porous plates with Ohmic dissipation and Heat Source." International J.Ultra Science, Vol. 18 November3(M) September- December 2006, pp.445-458.

3. P.R.Sharma and Bhawna Agarwal "Steady three-dimensional MHD flow along an infinite hot vertical porous surface bounded by a porous medium", J. Bulletin of pure and Applied Science. Vol. 25 E, 2006, pp.303-315. Accept in Mathematical Reviews (A division of the American Mathematical society, founded 1888) bibliographic files.

4. P.R.Sharma and Bhawna Agarwal "Steady MHD boundary layer flow and heat transfer along an infinite hot vertical porous moving plate with heat source." International J.Ultra Science, Vol. 18(1) M, Jan- April 2006, pp.123-130.

5. P.R.Sharma and Bhawna Agarwal "Steady laminar flow and heat transfer through parallel porous disc of different permeability with heat source." Indian J. of Theoretical physics, ISSN-0019-5693, VOL. 55, NO.2, 2007.

6. P.R.Sharma and Bhawna Agarwal, "Unsteady MHD flow and heat transfer along a vertical porous surface bounded by porous medium with Ohmic dissipation." Far East J. Appl. Math. 28(2), 2007, pp.265-81. Accept in Mathematical Reviews (A division of the American Mathematical society, founded 1888) bibliographic files.

TEXT AND REFERENCE BOOKS

1. Engineering mathematics –1, VAYU Publisher, New Delhi as per syllabus for B.Tech. 1st semester of various universities.

2. Bio statistical and Mathematics, Vardhan Publishers, Jaipur as per syllabus for biomedical students of Various Universities.

Dr. (Mrs.) Rajni Agarwal

Joined Institute on: 1-Jul-1993

Assistant Professor

Sciences

Ph.D.

CONTACT ADDRESS

Birla Institute of Technology, Mesra, Off-Campus, Jaipur

27, Malviya Industrial Area, Jaipur

Phone Office: 9784870160

Phone Residence: 9784870160

Email: ragarwal@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 17 Years

Research: 32 Years

Individual: 32 Years

PROFESSIONAL BACKGROUND

- Set up Engineering Chemistry Lab in 2004
- In-charge Cultural Program
- Co-Coordinator in Admission cell
- Assistant Warden of Girls Hostel
- Member Secretary in Environmental Protection Group
- Member in Anti-Ragging Squad Committee

SUBJECTS OF 2025

MO-25

- Chemistry
- Chemistry Lab

SP-25

- Constitution of India

RESEARCH AREAS

Water and Soil Pollution, Industrial waste water pollution

PUBLICATIONS

INTERNATIONAL PUBLICATIONS

1. Rajni Agarwal, "A Brief Review: Heavy Metals Pollutants in Drinking Water of Rajasthan, India", Poll Res. 45 (1): 159-163, 2026.

2. Rajni Agarwal, "A Study on Opportunities for Women & Girls in Entrepreneurship", Int. J. Sci. Res., 14 (6), 2025

3. Rajni Agarwal, "Nitrate Contamination in Ground Water of Jaipur District, Rajasthan, India: It's impact on Human Health: A Review", Int. J. Sci. Eng. & Res., 4 (11), 111-114, 2016.

4. Rajni Agarwal, "A Brief Review on Fluoride Concentration in Drinking Water in Malpura Tehsil (Tonk, Rajasthan, India), Bassi Tehsil (Jaipur, Rajasthan, India) and Vicinity Areas of Dausa District (Rajasthan, India)", Int. J. G. & H. Che., 3 (4), 1507-1512, 2014.

5. Rajni Agarwal, S. K. Sharma and Usha Jha, "Assessment of Heavy Metals in Industrial Effluents and in Ground Water of Sitapura Industrial Area, Jaipur (India) (During Pre-Monsoon Season2007)", Poll. Res., 30(3), 345-347, 2011.

6. Rajni Agarwal, S. K. Sharma and Usha Jha, "Assessment of soil quality of Sitapura Industrial Area, Jaipur during pre-monsoon season 2007", A. J. Che. & Env. Res., 4 (2), 64-66, 2011.

7. Rajni Agarwal, S. K. Sharma and Usha Jha, "Seasonal variation in the aquatic environment of Industrial effluents of Sitapura Industrial Area (pre-monsoon and monsoon season 2007)", Curr. W. Env., 4 (2), 385-388, 2009.

8. Rajni Agarwal, S. K. Sharma and Usha Jha, "Assessment of Industrial effluents and underground water during monsoon season 2007 in Sitapura Industrial Area, Jaipur", Curr. W. Env., 4 (1), 71-77, 2009.

NATIONAL JOURNALS

1. Rajni Agarwal, S. K. Sharma and Usha Jha, "Physico-chemical studies on Industrial effluents and ground water of Sitapura Industrial Area (SIA) in pre- monsoon season", Ind. J. Env. Sci., 13 (2), 161-165, 2009.

2. S. L. Mishra, S. K. Sharma and Rajni Agarwal, "Briquetting of Lignite for Domestic Fuel", J. Sci. & Ind. Res., 59, 413-416, 2000.

POSTER PRESENTED

1. Archna Shrivastava and Rajni Agarwal "Bioreclamation of Gypsum Mine-out Area" Proc. 92nd Ind. Sci. Cong. Part III.

ARTICLE PUBLISHED IN NEWSLETTER

1. "Conservation of Water is Today's Necessity" "VIPNET Curiosity" 3 (8), 7-9, OCTOBER 2022

RESEARCH PAPERS PRESENTED IN CONFERENCES

1. "To study the quality of ground water of Sitapura Industrial Area, Jaipur", "International Conference on Multi-disciplinary Perspective in Science & Humanities Recent Trends & Innovation (ICMPSH - 2025)" for 2 days 16th and 17th May 2025 in Arya Engineering College, Jaipur.

2. "Environmental Degradation and Treatment of Industrial Effluents", "National Conference on Environmental Conservation and Management for Sustainable Era [ECMSE} – 2012, S. S. Jain Subodh P. G. College, Jaipur – India, December, 20-22nd, 2012.

3. "Management of Industrial Effluents to Protect Environment", "National Conference on Water Crisis Management under Changing Climate", conducted by Gugly Centre for Biological Research, Bhubaneswar Odisha – India, November, 16-17th, 2012.

4. "Impact of Contaminated Water on Environment", "National Conference on Role of Green Chemistry for Risk Free Environment", RGCRFE-2010, Poornima Institute of Engg. & Technology, Jaipur – India, December 18th, 2010.

5. "Study of Significant Correlation Among Parameters of Industrial Effluents Samples during Pre-Monsoon Season of Sitapura Industrial Area, Jaipur (India)", "National Seminar on Role of Toxicants in Environmental Pollution: Causes, Effects and Control", RTEPCEC, Vedic Kanya PG College, Raja Park, Jaipur - India, February, 3-4th, 2010.

Dr. (Mrs.) Ranjana Agrawal

Joined Institute on: 1-Apr-2009

Assistant Professor

Chemistry

Ph.D.

CONTACT ADDRESS

Plot No. 306 Scheme 10-B, Gopalpura ByPass Road, Jaipur

Phone Office: 01414019815

Phone Residence: 9772204546

Email: ranjanaagrawal@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 21 Years

Research: 22 Years

Individual: 22 Years

PROFESSIONAL BACKGROUND

- Incharge Chemistry Department (2017 to 2022)
- Ex-Member of Examination Cell
- Coordinator - Student Induction Program
- Coordinator - R & D Cell (IIC)
- Member of IIC Cell
- Member Internal Complaint Committee
- Member of Disciplinary Committee
- Member of Anti Ragging Committee

SUBJECTS OF SP-25

MT131 UHV-II (BTECH-IV)

PROFESSIONAL PROFILES

LinkedIn: https://www.linkedin.com/in/ranjana-agrawal-7106226a/

ORCID: https://orcid.org/0009-0001-4857-9926

Google Scholar: https://scholar.google.com/citations?user=HPSdMzwAAAAJ&hl=en

RESEARCH AREAS

Water analysis and environmental pollution

AWARD AND HONOURS

Eminent guest and Speaker for a seminar held at Subodh PG girls college on "exploring millets; the chemistry nutritional value and its Economic impact" held on 6th April 2024

PUBLICATIONS

TOTAL PUBLICATIONS IN JOURNALS: 14

INTERNATIONAL JOURNALS (12):

1. Renu Dhetarwal, Kumud Tanwar, Ashok .K. Kakodia, Swati Singh, Siddhi Jain, Ranjana Agrawal

"Euphorbia Caducifolia: Applying Traditional Knowledge to Contemporary Pharmacology, A Comprehensive Review"

https://doi.org/10.52783/jchr.v15.i4.9467

2. Nidhi Singh, Swati Gupta, Ranjana Agarwal

"Bioaccumulation of Heavy Metals and Their Ecotoxicological Impacts on Selected Fish Species in Jaipur District Ponds"

Journal of Chemical Health Risks, www.jchr.org, JCHR (2025) 15(5), 1376-1383, ISSN:2251-672

3. Ranjana Agrawal, Rashmy Nair, Poonam Ojha

"Access of Flouride and Nitrate in ground water of Rajasthan- Their health hazards and remedial methods"

Ecology, Environment and Conservation Journal, May 2022, PP(S574-S576), ISSN 0971-765X

4. Ranjana Agrawal

"Contamination of Nitrate in ground water samples of Sawai Madhopur town Rajasthan"

International Journal of Green and Herbal Chemistry (Available online at www.ijghc.Com), Feb. 2013, Vol.2, No.1, 60-64

5. Ranjana Agrawal

"Fluoride contamination in ground water samples and its effect on human body in Gangapur city town (Sawai Madhopur District) Rajasthan"

Journal of Chemical, Biological and physical sciences, Feb. 2012-Apr. 2012, Vol. 2, NO.2, 1051-1053

6. Ranjana Agrawal

"Nitrate contamination in ground water samples of Gangapur city town (Sawai Madhopur District) Rajasthan"

Journal of Chemical, Biological and physical sciences, Nov. 2011-Jan. 2012, Vol. 2, NO.1, 511-513

7. Ranjana Agrawal

"Physico-chemical Analysis of selected Groundwater Samples of Behror Town in Alwar district, Rajasthan"

Asian Journal of Biochemical and Pharmaceutical Research, Issue 4, Vol.(1) 2011, 113-115

8. Ranjana Agrawal

"Physico Chemical characteristics of groundwater samples quality of Gangapur city town, Jaipur, Rajasthan"

Pollution Research Journal, Vol 29(4), 683-684, 2010

9. Ranjana Agrawal

"Volumetric determination of coenzyme Q 10 in pharmaceutical samples with ascorbic acid"

Journal of Chemistry and Chemical Sciences, Vol. 1, No. 1(2010), 21-28, December 2010

10. Ranjana Agrawal

"Physico Chemical Analysis of groundwater samples of Kotpulti town, Jaipur, Rajasthan"

International Journal of Chemical, Environmental and Pharmaceutical Research, Vol. 1, No. 2(2010), 111-113, Sept. 2010

11. Ranjana Agrawal

"An assessment of groundwater quality of Chaksu town in Rajasthan"

Current World Environment, Vol. 4, No. 1, Page 161-164, June 2010

12. Ranjana Agrawal

"Study of Physico-Chemical Parameters of Groundwater quality of DUDU town in Rajasthan"

Indian Journal of Pharmaceutical Sciences, Mumbai, Vol 2, No. 4 (2009), 969-971, Dec. 2009

NATIONAL JOURNALS (1):

1. Ranjana Agrawal

"An appraisal of ground water quality of Alwar town in Rajasthan"

Himalayan Journal of Environment Zoology, Vol 21, 157-161, 2007

CONFERENCES/SEMINARS/SYMPOSIUM/FDPs ATTENDED

Conferences attended: 18

Seminars attended: 11

Workshops attended: 5

Symposiums attended: 5

FDPs attended: 13

FACULTY DEVELOPMENT PROGRAMS

UHV-I, December 2023 (online)

UHV-II, 16-20 Jan 2024 (Lucknow, offline)

SEMINARS ATTENDED

Seminar on "Women's safety: Challenges and Initiatives" by Internal Complaint Committee BIT Mesra on 10th November 2025

POSTER PRESENTATIONS AND CONFERENCE PAPERS

1. Poster Presentation in RTES-2023 organised by SVNIT Surat held on 2nd and 3rd May 2023

Title: "Analysis of concentration of Nitrate and Fluoride in drinking water of Kotputli and Phagi city of Jaipur district"

Authors: Dr. Ranjana Agrawal, Dr. Bhawana Agarwal, Mrs. Nidhi Singh

2. Paper Presentation in three days International Conference on "Advances in Materials and Chemical Sciences" held on 18-20 Dec 2024 in BIT Mesra, Ranchi

Title: "Cadmium Sulphide nanoparticle; their Synthesis and Application"

Authors: Dr. Ranjana Agrawal, Dr. Swati Singh, Dr. Kumud Tanwar, and Siddhi Jain

ORGANIZER

1. Summer Workshop on "Waste Water Treatment and Protection of Water Resources" from 5th May 2025 to 17th May 2025

2. Summer Training Program on "Emerging issues in Water Pollution, AI-based Monitoring and Remediation Strategies" held from 15th May to 25th May 2025

MEMBER OF PROFESSIONAL BODIES

- Life Membership of Indian Science Congress
- Life Membership of Indian Society of Analytical Scientists

Dr. Dheeraj Bhardwaj

Joined Institute on: 15-Sep-2010

Assistant Professor

Physics

Ph.D., M.Sc.

CONTACT ADDRESS

Nemi Nagar Extn., Vaishali Nagar, Jaipur-302021, Rajasthan

Phone Office: 8949721312

Phone Residence: 8949721312

Email: dbhardwaj@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 24 Years

Research: 20 Years

Individual: 15 Years

PROFESSIONAL BACKGROUND

PROFESSIONAL QUALIFICATIONS

Ph.D. (Microstrip Antennas) from University of Rajasthan, Jaipur in 2011.

MSc. (Physics) with Specialization in Microwave Electronics from University of Rajasthan, Jaipur in 2001

B.Sc. (PCM) Electronics from University of Rajasthan, Jaipur in 1998.

CURRENT SEMESTER SUBJECTS (SP25)

- PH24101 Physics (B.Tech (CSE)-II Sem)
- PH24102 Physics Lab (B.Tech (CSE)-II Sem)
- PH24102 Physics Lab (B.Tech (ECE)-II Sem)
- PH24102 Physics Lab (B.Tech (EEE)-II Sem)
- PH113 Physics (B.Tech -SEM VIII SP25-2021-2025 Section-Additional - I)

OFFICIAL RESPONSIBILITIES

- Hindi Adhikari, BIT Jaipur Campus (2020 to till date)
- Faculty mentor for Antenna Research Club (2020 to till date)
- Member of News and Publication cell as Hindi Adhikari (2024 to till date)
- Member of Research and Development Cell (2025 to till date)

- Member of Canteen Committee (2011 to till date)
- Member of Anti Ragging Squad (2024 to till date)
- Member of Internal Complaint Committee (2022 to till date)
- Ex-In-charge- Department of Physics, BIT Jaipur Campus (2011 to 2022)
- Ex-Faculty In-charge- Student Registration BIT Jaipur Campus (2021 to 2022)
- Ex-Faculty In-charge- Administration BIT Jaipur Campus (2021 to 2022)
- Ex-In-Charge, Department of Physics, BIT Jaipur Campus (2011 to 2022)
- Ex-Member of Doctoral Committees, BIT Jaipur Campus (2011 to 2019)
- Ex-Member of Staff Quarter allotment Committee, BIT Jaipur Campus (2011 to 2023)
- Member of Application form Scrutiny Committee for Teaching and Non-teaching staff (2021-2022)

RESEARCH AREAS

Defected Ground Structures (DGS) integration technique for Microstrip Planar Antennas, New Modes and Feeding Techniques for Antennas, Wideband Antenna using Hybrid/Composite Configurations High Gain Antennas for Wireless Communication.

AWARD AND HONOURS

AWARDS & RECOGNITIONS

Chaired Sessions at International Conferences

1. Chaired a Session at International Conference: "Recent Trends in Emerging Technologies [iNCRTET-2022]". This conference was organized by the Department of Physics, SKIT M&G, Jaipur in collaboration with the Innovative Technology Enabling Centre (InTEC) of the CSIR-Institute of Minerals and Materials Technology (IMMT) on March 16-17, 2022.

2. Chaired a Session at International Conference: "Advancements in Nano-electronics and Communication Technologies" (ICANCT-2020). This conference was organized by the Department of ECE, SKIT M&G, Jaipur. Session Details: Related to Antennas, Session Date: 17.01.2020 (Friday), Time: 3:00 pm - 5:00 pm, Venue: SKIT Jagatpura.

ACADEMIC & RESEARCH CONTRIBUTIONS

1. Appointed as an external examiner for viva-voce examination of Doctoral Research Scholar Mr. Parvez Ansari (Vivekananda Global University, Jaipur) for the award of Ph.D. Degree guided by Professor (Dr.) Yogesh Sharma Dean, Vivekananda Global University Jaipur. The viva-voce examination was conducted at 09:30 AM on 25 March, 2021. The Title of thesis was "Development of Chalcogenide Multilayer Thin Film Structures for Solar Cell Devices".

2. Appointed as an external examiner for M.Tech. (Information and Communication Technology) final-year students at the University of Rajasthan, Jaipur (2021).

3. Served as a reviewer of research papers for Defense Science Journal (SCIE Journal) published by DRDO, New Delhi (2020 to till date).

4. Served as a reviewer for an SCIE Journal at Taylor's University, Malaysia (2017 to till date).

5. Appointed as an external examiner. Prepared the M.Tech. End Semester question paper on Antenna Theory and Techniques for Jagannath University, Jaipur (2016).

6. Conducted the Viva Voce examination for M.Sc. Physics Semester-IV students at The IIS University, Jaipur (2016).

PUBLIC SPEAKING & RECOGNITION

1. Delivered a talk through All INDIA RADIO Jaipur on National Technology Day via All India Radio, Jaipur (2020). The Talk was Aired through All India Radio Jaipur on the Occasion of Celebration of National Technology Day from 7AM to 7:20AM, 11 May 2020.

2. Secured first place for research poster presentations at International conferences at IIS University, Jaipur (2015) and SKIT, Jagatpura, Jaipur (2013).

PUBLICATIONS

LIST OF PATENTS APPLIED/PUBLISHED

1. Dr. Dheeraj Bhardwaj, Department of Physics, et al., Applied for an Indian Patent on the title of "A Smart Sun Tracking Solar Panels" Application No.- 358873-001 Applied on the date 20/02/22.

2. Dr. Dheeraj Bhardwaj, Department of Physics, et al., Applied for an Indian Patent on the title of "Novel 5G communication microstrip patch antenna for broadcasting and navigation application." Application No.- 202241051840 Applied on the date 11/09/2022, Publication Date 16/09/2022.

3. Dr. Dheeraj Bhardwaj, Department of Sciences (Physics), et al., has published an Indian Patent as a fifth Inventor on the title of "Design of Microstrip Patch Antenna Inspired by Spiritual Exotic Om Shape for Satellite Communications." Application No.- 202411068116, Application date: 10/9/24, Publication Date 22/11/24.

INTERNATIONAL JOURNAL PUBLICATIONS

1. R. Kumar, Y. Chandra Sharma, V. V. Sagar and D. Bhardwaj, "Characterization of Low Temperature Plasma Ion Nitriding (PIN) of Inconel 600 and 601 Alloys" Iranian Journal of Materials Science & Engineering, Vol. 17, No. 2, June 2020. (ESCI & Scopus Indexed)

2. Dheeraj Bhardwaj, Komal Sharma and Satyarth Sharma, "Bandwidth Enhancement of Circular Microstrip Patch With Phi Shape Gap Coupling Slot By Varying Air Gap" Vol. 34 (n. 9, 2019). ISSN:2416-3953, Publication Date: 25 December 2019. (SCIE Journal)

3. Dheeraj Bhardwaj and Komal Sharma, "Dual Frequency Dual Band OM Shape Microstrip Patch Antenna for Wi-Max Applications" Vol. 34 (n. 9, 2019). ISSN:2416-3953, Publication Date: 25 December 2019. (SCIE Journal)

4. Kanika Joshi, Dr. Dheeraj Bhardwaj, Dr. Vivekanand Tiwari, "On radiation from convexo-concave microstrip patch antenna with impedance matching transformer," Volume 7 Issue 1, pp 731-735- May 2016, DOI : http://dx.doi.org/10.21172/1.71.104, Impact Factor 0.685

5. Deepak Verma, Dheeraj Bhardwaj, Kanika Joshi, Komal Sharma, "Review Paper on Modified Rectangular Shaped, Microstrip Patch Antennas", International Journal of Electronics & Communication Technology, IJECT Vol. 7, Issue 2, April - June 2016. Impact Factor 0.675, U.G.C. Approved Journal- S.No.-3918.

6. Dheeraj Bhardwaj, Sahil, Komal Sharma, "Modified stacked compact microstrip patch antenna for implantable medical devices", International Journal of Latest Trends in Engineering and Technology (IJLTET), vol 1, Issue 7, pp- 724-730, 2016. Impact Factor 0.685

7. Dheeraj Bhardwaj, Srishti Saraswat, Gitansh Gulati, Komal Sharma, "Design of defected ground low pass microstrip filters with enhanced performance characteristics", International Journal of Latest Trends in Engineering and Technology (IJLTET), vol 1, Issue 7, pp- 736-741, 2016, Impact Factor 0.685.

8. Gitansh Gulati, Dheeraj Bhardwaj, Manan Dixit, Komal Sharma, "Gain and Bandwidth Enhancement of Rectangular Patch Antenna with Trapezoidal Slot for Wi-Max Applications" International Journal of Engineering Research and Technology, Vol.3, Issue 11, pp.1072-1076, November-2014, e-ISSN:2278-0181. Impact Factor 2.8, U.G.C. Approved Journal - S.No.-5060.

9. D. Bhardwaj, O. P. Sharma, C. K. Dubey, B.V. Singh, and Komal Sharma, "Gain and Bandwidth Enhancement of Electromagnetic Gap-Coupled Assembly of Various Patches Forming Rhombus Shaped Microstrip Patch Antenna for C-Band Applications" Int. Journal of Engineering Research and Applications, ISSN : 2248-9622, Vol. 4, Issue 5(Version 1), May 2014, [IJERA Impact Factor 5.179], U.G.C. Approved Journal - S.No.-4522.

10. D.Bhardwaj, O. P. Sharma, B.V. Singh, C. K. Dubey and Komal Sharma, "Broadband Electromagnetic Gap-Coupled Assembly of Patches Forming Trapezoidal Microstrip Antenna for S-Band Application" Int. Journal of Engineering Research and Applications, ISSN : 2248-9622, Vol. 4, Issue 5(Version 1), May 2014, [IJERA Impact Factor 5.179], U.G. C. Approved Journal -S.No.-4522.

11. D.Bhardwaj, Rakshit Jain, Vijay Sharma and Komal Sharma, "Broadband Gap-Coupled Assembly of Patches Forming Elliptical Microstrip Patch Antenna for C- Band Applications", International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering, ISSN (Online) 2321 – 2004, Vol. 2, Issue 5, May 2014. [Impact Factor 5.359], U.G. C. Approved Journal - S.No.-4799.

12. D.Bhardwaj, Nidhi jain, Komal Sharma, "Bandwidth Enhancement of Broadband Dual Resonator I-Shaped Antenna for C-Band Applications", International Journal of Electronics & Communication Technology (IJECT), ISSN : 2230-9543 (Online) Vol. 5, Issue 1, Jan - March 2014. [Globle Impact factor 0.675], U.G.C.- Approved Journal S.No.-3918.

13. D.Bhardwaj, D Verma, Komal Sharma, "Design of dual band broadband modified rectangular microstrip antenna with air gap for wireless applications", International Journal of Enhanced Research in Science Technology & Engineering, Vol.3, Issue2, pp: (331-339), February-2014. ISSN: 2319-7463, [Impact Factor: 4.059]

14. D.Bhardwaj, Komal Sharma, "Broadband rectangular patch microstrip antenna with rhombus shaped slot for WLAN applications", International Journal of Enhanced Research in Science Technology & Engineering, ISSN: 2319-7463 Vol. 3 Issue 1, pp: (389-394), January-2014. ISSN: 2319-7463, [Impact Factor: 4.059]

15. D.Bhardwaj, Komal Sharma and Deepak Bhatnagar "Dual band and Broadband Rectangular Patch Mircostrip Antenna with T Shaped Slot for WiMax Application", International Journal of Engineering Research and Development, PISSN: 2278-800X, Volume 3, Issue 12, PP. 14-21, September 2012. [Globle Impact factor 1.458], U.G.C. Approved Journal- S.No.-4739.

16. Dheeraj Bhardwaj, D. Bhatnagar and S. Sancheti "Design Patch Antenna with a Notch on FR4 Substrate", Journal of IET Microwave Antennas Propagation, Vol. 2, No. 8, pp. 880–885, 2008. ISSN : 1751-8725. Impact factor 1.297, SCI JOURNAL.

17. Dheeraj Bhardwaj, D. Bhatnagar, S. Sancheti and Brijesh Soni, "Radiations from double notched square patch antenna on FR4 substrate" Journal of Microwaves, Optoelectronics and Electromagnetic Applications, Vol. 7, No.2, December 2008. ISSN 1516-7399. Scopus Index Journal.

INDIAN JOURNAL PUBLICATIONS

1. Dheeraj Bhardwaj, Maninder Singh Lakha, Komal Sharma, "Bandwidth Enhancement of Modified Rectangular Microstrip Antenna With W Shaped Slot For Wi-Max Application" SKIT Research Journal, Vol 4; Pp-78-81, ISSUE 1: 2014

2. Brajraj Sharma, Komal Sharma, Vijay Sharma, Dheeraj Bhardwaj, KB Sharma and D Bhatnagar "Circularly polarized K shaped Planar Antenna for Wi Max Applications" SKIT Research Journal, ISSN2278-2508, pp 178-180, Vol-3, 2013.

3. Komal Sharma, Dheeraj Bhardwaj and Brajraj Sharma, "Dual band square Microstrip Patch Antenna with two Rectangular Slots for Wi-Max Applications" SKIT Research Journal, ISSN2278-2508, pp 178-180, Vol-3, 2013.

4. Komal Sharma, Dheeraj Bhardwaj and Brajraj Sharma, "Rectangular Patch Microstrip antenna with U Shaped Slot with Shorted Pin and Walls for WiMax and WLAN Application" SKIT Research Journal, ISSN2278-2508, Vol-2, 2012.

5. Dheeraj Bhardwaj, Komal Sharma, D. Bhatnagar and S.Sancheti "Compact Parasitically coupled concentric semi circular elliptically ring antenna surrounding an elliptical patch", Indian Journal of Radio and Space Physics, Vol. 39, April 2010. Scopus Index Journal. UGC Approved Journal-S.No.-1806.

SPRINGER BOOK CHAPTERS (2)

1. Dheeraj Bhardwaj, Gitansh Gulati, Lakshit Bhardwaj and Komal Sharma, "Gain and Bandwidth Enhancement of Single Layered Slotted Triangular Shaped Microstrip Patch Antenna with Stub for WLAN Applications", published in Springer special issue, pp 729-739, Dated : 29 April 2016.

2. Dheeraj Bhardwaj, Aayush Dwivedi, Nidhi Jain, Komal Sharma "Design of Dual Resonator Broadband Multilayer Electromagnetically Coupled Microstrip Antenna for X-Band Applications" published in Springer special issue, pp 741-750, Dated: 29 April 2016.

AMERICAN INSTITUTE OF PHYSICS (AIP) PUBLICATIONS (2)

1. Dheeraj Bhardwaj, Gitansh Gulati, Srishti Saraswat and Komal Sharma, "Bandwidth Enhancement of Electromagnetic Coupled Non-uniform H-Shaped Microstrip Patch Antenna for Higher Band of Wi-MAX Applications", Scopus Indexed AIP Conference Proceedings 1715 (1), 020033. 2016.

2. Dheeraj Bhardwaj, Gitansh Gulati, Srishti Saraswat and Komal Sharma, "Dual Band Multi Frequency Rectangular Patch Microstrip Antenna with Flyswatter Shaped Slot for Wireless Systems", Scopus Indexed AIP Conference Proceedings 1715 (1), 020044. 2016.

INTERNATIONAL CONFERENCE PUBLICATIONS

1. Dheeraj Bhardwaj, Deepak Verma and Komal Sharma, "Design of Modified Rectangular Shaped Slotted Antenna for Wireless Applications" been accepted for presentation at the 1st International Conference on Innovative Advancements in Engineering and Technology (IAET), organized by Department of Electronics and Communication Engineering, Jaipur National University, Jaipur, March 7th and 8th, 2014.

2. Saksham Soni, Vidisha Garg, Vipul Jain, Dheeraj Bhardwaj and Komal Sharma "Dual Band Elliptical Microstrip Patch Antenna with Triangular Slot For WLAN Application" In Proceedings of International

Conference on "Computational Electronics and Nanotechnology" held at AMITY University Rajasthan, Jaipur, 1st and 2nd March 2012.

3. Richa Dolia, Tanu Rawat, Dheeraj Bhardwaj and Komal Sharma, "Compact Triangular Microstrip Antenna with Triangular Slot for Bluetooth Application" In Proceedings of International Conference on "Computational Electronics and Nanotechnology" held at AMITY University Rajasthan, Jaipur, 1st and 2nd March 2012.

4. Komal Sharma, Dheeraj Bhardwaj, and Deepak. Bhatnagar, "Dual frequency broadband rectangular patch mircostrip antenna with star shaped slot for Wi-Max application" In Proceedings of International conference on "Adaptive computer technologies in various engineering applications" Poornima college of engineering, Jaipur, 24th to 26th February, 2011.

5. Dheeraj Bhardwaj, Komal Sharma and D. Bhatnagar "Multi frequency wide band pin loaded rectangular microstrip patch antenna with y shaped slot for future generation communication systems", In Proceedings of International conference on "Adaptive computer technologies in various engineering applications", Poornima college of engineering, Jaipur, 24th to 26th February, 2011.

6. Komal Sharma, Dheeraj Bhardwaj, D. Bhatnagar, and S. Sancheti "Shorting pin loaded multi frequency broadband rectangular patch mircostrip antenna with star shaped slot for wimax application" International Conference on "Emerging Interfaces of Physical Sciences", University of Rajasthan, Jaipur, 22nd to 24th December, 2010.

7. Dheeraj Bhardwaj, Komal Sharma, D.Bhatnagar, S. Sancheti and Brijesh Soni, "Comparison between different Elliptical Patch Microstrip Antenna with Various Slots", 12th International Symposium on Microwave and Optical Technology (ISMOT-2009), Jodhpur. 19th to 21st December 2009.

8. Dheeraj Bhardwaj, Komal Sharma, D. Bhatnagar and S.Sancheti, "Electrically Thick Parasitically Coupled Concentric Semi Circular Elliptically Ring Antenna Surrounding an Elliptical Patch", In Proceedings of International Conference on Microwaves, Antenna, Propagation & Remote Sensing, "ICMARS-2009", Jodhpur. 19th to 21st December 2009.

9. Dheeraj Bhardwaj, Komal Sharma, D. Bhatnagar and S.Sancheti, "Broad Band Parasitically Coupled Concentric Semi Circular Elliptically Ring Antenna Surrounding an Elliptical Patch with Air Gap", In Proceedings of IEEE Applied Electromagnetic Conference AEMC-09 Kolkata, 14th to 16th December 2009.

10. Dheeraj Bhardwaj, Komal, D.Bhatnagar and S.Sancheti "Square patch microstrip antenna with single notch having an air gap", In Proceedings of International Conference on Recent Advancements in Microwave Theory & Applications, University of Rajasthan Jaipur, November 21-24, 2008.

11. Dheeraj Bhardwaj, V. Bhardwaj, Komal, D. Bhatnagar and S. Sancheti "Radiation performance of an elliptical patch antenna with a circular slot for modern communication systems", In Proceedings of International Conference on Recent Advancements in Microwave Theory & Applications, University of Rajasthan Jaipur, November 21-24, 2008.

NATIONAL CONFERENCE/SYMPOSIUM/SEMINAR PUBLICATIONS

1. Dheeraj Bhardwaj, Nidhi Jain and Komal Sharma, "Comparison of Simulated Radiation Performance of Uniform and Nonuniform H-Shaped Microstrip Patch Antennas Using IE3D Simulation Software", National Conference on Emerging Computing Technologies in modern ERA, held at S.S. Jain Subodh College on 22-23, January 2014.

2. Dheeraj Bhardwaj, Deepak Verma, Snehanshu Shekhar and Komal Sharma "Design of Modified Broadband Microstrip Rectangular Shaped Slotted Antenna For WLAN Application" National held at S.S. Jain Subodh College on 22-23, January 2014.

3. Dheeraj Bhardwaj, Meenakshi Sharma, Snehanshu Shekhar and Komal Sharma "Broadband Triangular Patch Microstrip Antenna with Two Rectangular Slots for WLAN Application", National Conference on Emerging Computing Technologies in modern ERA, held at S.S. Jain Subodh College on 22-23, January 2014.

4. Dheeraj Bhardwaj, Gaurav Goyal and Komal Sharma "Dual band Rectangular Microstrip Patch Antenna with Hexagonal and Y Shaped Slot for X-Band and C-Band Application" National Conference on Emerging Computing Technologies in modern ERA, held at S.S. Jain Subodh College on 22-23, January 2014.

5. Dheeraj Bhardwaj, Nidhi Jain and Komal Sharma, "Broadband Dual Resonator Uniform H-Shaped Microstrip Patch Antenna for C-Band Applications" National Conference on "Advancement in wireless & optical communication, held at Poornima College of Engireering, Jaipur, March 18-19, January 2014.

6. Dheeraj Bhardwaj, Manan Dixit and Komal Sharma, "Broadband Gap-Coupled Rectangular Microstrip Patch Antenna with Trapezium Slot for Wi-Max Application" National Conference on "Energy, Materials & Communication Systems, Rajdhani Engg. College, Jaipur, March 16-17, 2013.

7. Dheeraj Bhardwaj, Nishita Modi, Meenal Saini, Brajraj Sharma and Komal Sharma "On Radiation form Rhombus Shaped Mircostrip Patch Antenna with Rhombus Slot for Wi-Max Application" National conference on "advancements in microelectronics and communication technologies, SKIT Engg. College Jaipur. March 15-17, 2013.

8. Dheeraj Bhardwaj, Namrata Sinha, Paridhi Jain, Rachita Agrawal, Brajraj Sharma and Komal Sharma "Broadband Parallelogram Microstrip Patch Antenna with Rectangular Slot for Wi-Max Application" National conference on "advancements in microelectronics and communication technologies, SKIT Engg. College Jaipur. March 15-17, 2013.

9. Kapil sharma, Shubham Kumar, Dheeraj Bhardwaj and Komal Sharma, "Effect of Substrate Thickness on The Performance of Circular Microstrip Patch Antenna With Elliptical Slot For Wimax Application", National conference on "Micrwave Device and Components", held at Global Institute of Technology, Sitapura, Jaipur, 22nd-23rd March-2012.

10. Saksham Soni, Vidisha Garg, Vipul Jain, Dheeraj Bhardwaj and Komal Sharma, "Dual Frequency Rectangular Microstrip Patch Antenna with Diagonal Square Slots for Wireless Communication", National conference on "Microwave Device and Components", held at Global Institute of Technology, Sitapura, Jaipur, 22nd-23rd March-2012.

11. Saksham Soni, Vidisha Garg, Vipul Jain, Dheeraj Bhardwaj and Komal Sharma, "On Radiation from Dual Band Rectangular Microstrip Patch Antenna with Rectangular Slot", National conference on "Advances in Wireless & Optical Communication Systems" (AWOCS-2012), held at Swami Keshvanand Institute of Technology, Management & Gramothan (SKITMG), Jagatpura, Jaipur, 17th-18th March-2012.

12. Saket Suman, Ritika Kapoor, Nehal Gupta, Shruti Mehrotra, Dheeraj Bhardwaj, Komal Sharma and Deepak Bhatnagar "Dual Frequency Dual Band Rectangular Patch Microstrip Antenna with U Shaped Slot for WiMax Application, RCERT, Jaipur, 26th-27th March-2011.

13. Komal Sharma, Dheeraj Bhardwaj and D. Bhatnagar "Multi frequency Wide Band Modified Square Patch Antenna for Next Generation Communication Systems National Conference, Agra, 26th-27th March-2011.

14. Komal Sharma, Dheeraj Bhardwaj, D. Bhatnagar, and S. Sancheti, "Compact dual frequency broad band square patch with single notch microstrip antenna using shorting pin and shorting wall technique for wlan application", National Conference on "Recent Advances in Modern Communication Systems and Nanotechnology NCMCN-2011" held at University of Rajasthan, Jaipur, on 06th to 08th January, 2011.

15. Komal Sharma, Dheeraj Bhardwaj, D. Bhatnagar, and S. Sancheti "Multi Frequency Triple Band Rectangular Patch Mircostrip Antenna with T Shaped Slot for WiMax Application", National Conference on "Recent Advances in Modern Communication Systems and Nanotechnology NCMCN-2011" held at University of Rajasthan, Jaipur, on 06th to 08th January, 2011.

16. Komal Sharma, Dheeraj Bhardwaj, D. Bhatnagar S. Sancheti, "Design of Multi Frequency Broadband Rectangular Patch Mircostrip Antenna with Y Shaped Slot", national Conference on "Advances in Wireless and Optical Communication Technologies", Poornima College of Engineering, Jaipur, 7th to 8th August 2010.

17. Dheeraj Bhardwaj, Komal Sharma, D. Bhatnagar and S. Sancheti "Radiations from Elliptical patch Microstrip Antenna with and without Circular Slot", National Conference on "Advances in communication technology in cyber Age", IETE Jaipur, 17th May 2009.

MEMBER OF PROFESSIONAL BODIES

- Life Member of Indian Physics Association [I.P.A.]
- Life Member of Indian Association of Physics Teachers [I.A.P.T.]
- Life Member of Rajasthan Physics Association [R.P.A.]

Dr. Indra K. Dadheech

Joined Institute on: 2-Jan-2006

Assistant Professor

Applied Sciences

Ph.D. (University of Rajasthan), M.Sc. (University of Rajasthan)

CONTACT ADDRESS

72, Bajaj Nagar Enclave, Jaipur - 302015 Rajasthan

Phone Office: 0141 4019813

Phone Residence: 0141 4019816

Email: ikdadheech@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 19 Years

Research: 9 Years

Individual: 10 Years

PROFESSIONAL BACKGROUND

COURSES TAUGHT

- Mathematics I (MA24101)
- Mathematics II (MA24103)
- Numerical Methods along with Subject Lab (MA203 and MA204)
- Discrete Mathematics
- Graph Theory

OFFICIAL RESPONSIBILITY

- In-Charge Examination

PROFESSIONAL PROFILES

LinkedIn: https://www.linkedin.com/in/inder-dadheech-573585220/

Google Scholar: https://scholar.google.com/citations?view_op=search_authors&mauthors=I+K+Dadheech&hl=en&oi=ao

RESEARCH AREAS

Fluid Dynamics

PUBLICATIONS

1. "Effect of volumetric heat generation / absorption on convective heat and mass transfer in porous medium between two vertical porous plates". International Journal of Engineering Research and Technology, Vol. 1, Issue 3, 2012, ISSN: 2278-0181.

2. "Effect of Joule heating on steady MHD flow of low Prandtl fluid on a porous stretching sheet". International Journal of Engineering Research and Technology, Vol. 1, Issue 3, 2012, ISSN: 2278-0181.

3. "Heat and mass transfer effects on unsteady MHD free convective flow along a vertical porous plate with internal heat generation and variable suction". International Journal of Mathematical Archive, Vol. 3, Issue 5, 2012, ISSN: 2229-5046.

4. "Effects of variable thermal conductivity and internal heat generation on unsteady MHD flow through porous media bounded by a vertical porous plate with variable suction". International Journal of Mathematical Modeling Simulation and Application, 2012.

WORKSHOP/CONFERENCE ATTENDED

1. Two week ISTE Workshop on "Fluid Mechanics" from May 20-30, 2014, conducted by IIT, Kharagpur under National Mission on Education through ICT (MHRD)

2. Conference on "Harmonic Analysis and Operator Theory" from March 21 – 23, 2014, organized by Department of Mathematics, IISER, Bhopal

3. UGC sponsored DRS-I under SAP-National Workshop on "Challenges before Applied Mathematics: Fluid Dynamics and Optimization Techniques" from March 11-13, 2011, organized by Department of Mathematics, University of Rajasthan, Jaipur.

4. National Seminar on the topic "Recent developments in Fluid Dynamics and 14th Prof. P. D. Verma memorial lecture" from August 7-8, 2010 at University of Rajasthan conducted by Department of Mathematics, University of Rajasthan, Jaipur.

5. National Seminar on the topic "Recent trends in mathematics and actuarial sciences" from February 20-21, 2010 at Malvia National Institute of Technology (MNIT) Jaipur, conducted by National University of Rajasthan.

6. National Seminar on the topic "Computational mathematics and operation reearch" from November 28-29, 2009 at B. K. Birla Institute of Engineering and Technology, Pilani, Rajasthan.

7. AICTE sponsored staff Development Program entitled "Basics and application of computational fluid dynamics" from July10-23, 2009, conducted by Department of Mechanical Engineering, Thapar University, Patiala (Punjab)

8. Quality Improvement Programme on the topic "Computational Fluid Dynamics" from June 16-27, 2008 at IIT Roorkee.

9. Quality Improvement Programme on the topic "Numerical Computation Using Software Tools" from July 14-18, 2007 at IIT Roorkee.

10. Faculty Development Programme on the topic "Application Orientation in Engineering Mathematics" from June 25 to July 06, 2007 at Malvia National Institute of Technology, Jaipur.

11. Indo-Australian Workshop on the topic "CFD Approach on Fluid Flow, Heat and Mass Transfer" from April 12-14, 2007 at IIT Roorkee.

12. Short term Course on the topic "Mathematical Methods in Engineering and Science" from July 03-15, 2006 at IIT Kanpur conducted by Centre for Development of Technical Education.

13. National Conference on the topic "Science and technology for Desert Development" from December 02-04, 2004 at BISR, Jaipur.

LECTURE DELIVERED/PRESENTATION

1. Delivered an expert lecture on "Partial differential equation and solutions" in AICTE sponsored staff Development Program entitled "Basics and application of computational fluid dynamics" conducted by Department of Mechanical Engineering, Thapar University, Patiala (Punjab).

2. Presented a paper entitled "Effects of heat generation/absorption and chemical reaction on flow of water at 40C along a continuously moving porous vertical plate" in National Conference on the topic "Mathematical modeling and simulation" conducted by Department of Mathematics, Guru Jambeshwar University of Science and Technology, Hissar, Haryana.

3. Presented a paper entitled "Free convective flow and mass transfer through a porous medium bounded by infinite vertical porous plate in the presence of volumetric heat generation/absorption" in UGC sponsored National Symposium on the topic "Application of various techniques in Fluid dynamics" conducted by Department of Mathematics B.S.N.V PG College, (Luknow University) Luknow in association with National Science Network

4. Presented a paper entitled "Effect of volumetric heat generation / absorption on convective heat and mass transfer in porous medium between two vertical porous plates" in National seminar on "Recent trends in fluid dynamics and optimization techniques", conducted by Department of Mathematics, University of Rajasthan, Jaipur.

WORKSHOP ORGANIZED

Organized a one week winter school on "Introduction to Linear Algebra and its Application" from December 17-21, 2013 for engineering students at BIT, Mesra (Ranchi), Jaipur Campus

MEMBER OF PROFESSIONAL BODIES

- Life member of 'Rajasthan Ganitha Prishad', Ajmer, Rajasthan.
- Life member of 'Indian Society of Theoretical and Applied Mechanics', IIT, Kharagpur.

TEXT AND REFERENCE BOOKS

Write three book chapters for Vardhmaan Mahaveer Open University for the course "Numerical Analysis", (MA/MSc. MT- 08).

Dr. Raj Kumar Gothwal

Joined Institute on: 1-Aug-2009

Assistant Professor

Sciences

Ph.D.

CONTACT ADDRESS

Department of Bioengineering and Biotechnology, BIT Mesra, Jaipur-Campus, 27, Malviya Industrial Area, Jaipur-302017

Phone Office: 0141-2385283

Phone Residence: 9414778735

Email: rkgothwal@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 17 Years

Research: 26 Years

Individual: 26 Years

PROFESSIONAL BACKGROUND

SUBJECTS TAUGHT

- Cell and Molecular Biology (BI101)
- Proteomics, Metabolomics & Biomarker Design (BI301)
- Molecular Biology and rDNA Technology (BT415)
- Genomics (BT407)
- Biological Sciences (BT3021)
- Biological Sciences (BT3009)
- Biological Science for Engineers (BE101)
- Biological Science for Engineers (BE24101)

RESEARCH AREAS

Applied Microbiology, Microbial Enzymes, Plant-Microbial Interactions, Microbial Bioprospecting, Medical Biotechnology

PROFESSIONAL PROFILES

Scopus Profile: https://www.scopus.com/authid/detail.uri?authorId=12242620500

Google Scholar: https://scholar.google.co.in/citations?user=BEyYELsAAAAJ

Web of Science: https://www.webofscience.com/wos/author/record/JDW-8308-2023

AWARD AND HONOURS

1. Chaired a session in the DST sponsored International Conference on Cells to Communities, Interlinking life science, Forestry and Social Development at IIS University, Jaipur during February 4-6, 2026.

2. Extern Subject Expert - Board of Studies (BoS) in JNU, Jaipur. Meeting held on June 03, 2023 organized at JNU, Jaipur.

3. Judge - In the event EXPRESSIONS (Science oriented technical skit) in Annual Festival 'Technorazz2022' organized at School of Life Sciences, Jaipur National University during September 30 – October 1, 2022.

4. Chair - International Conference on Biotechnology for Sustainable Agriculture, Environment and Health (BASEH-20021) organized at Birla Institute of Scientific Research, Jaipur during April 4-8, 2021.

5. Centre Observer – Appointed on behalf of Government of Rajasthan to conduct Rajasthan Eligibility Exam for Teachers (REET) at SKIT Campus, Jaipur, Rajasthan during September 26, 2021.

6. Chair - National Conference on Biodiversity Conservation for a Sustainable Future at IIS University, Jaipur during October 11-12, 2019.

7. Rapporteur - International Conference, INBIX'19, Indian Conference on Bioinformatics at HRMMV, Jalandhar, Punjab during February, 22-23, 2019.

8. Chair – Frontiers in Genetics: From Laboratory to Therapeutics, Systems Genetics and Therapeutic Biology, Symposium organized at Birla Institute of Scientific Research, Jaipur on September 29, 2018

9. Judge - Paper Presentation, Annual Festival 'Technorazz2017' organized at School of Life Sciences, Jaipur National University on November 10, 2017.

10. Platinum Jubilee Best Poster Award - 54th Annual Conference of Association of Microbiologists of India organized at M. D. University, Rohtak, India during November 17-20, 2013.

11. Merit Scholarship – Department merit scholarship, Department of Biochemistry, Rajasthan College of Agriculture, Udaipur during 1991-1993.

PUBLICATIONS

RESEARCH SUPERVISION

PhD: 1 (Awarded), 6 (Ongoing)

MTech: 9

MSc: 66

BTech: 36

BSc: 8

RESEARCH PUBLICATIONS (JOURNAL PAPERS)

1. Monisha Singhal M, Chaturvedi P, Gothwal R K, Mohan M K & Solanki P S (2021). Comparative study of synonymous codon usage in bacteria growing at extreme temperatures. Int. J. Bioinfo. Res. Appl., 17(1): 53-68. DOI: 10.1504/IJBRA.2021.113966.

2. Nehra S, Gothwal R K, Meena P, Ghosh P & Trivedi P C (2020). Fungal diseases associated with coconut and their management - A review. J. Ind. Bot. Soc. – Special Centenary Volume, 100 (A): 239-250.

3. Pradhan S., Gothwal R. K., Mohan M. K. and Ghosh P. (2019). Evaluating bacterial cell immobilization of Brevibacillus formosus BISR-1 and Paenibacillus sp. BISR-047 with different matrices. Int. J. Engg. Res. Technol., 8(6): 1358-1363. DOI: 10.17577/IJERTV8IS060713.

4. Meena S., Gothwal R. K., Saxena J., Nehra S., Mohan M. K. and Ghosh P. (2015). Effect of metal ions and chemical compounds on chitinase produced by a newly isolated thermotolerant Paenibacillus sp. BISR-047 and its shelf-life. Int. J. Curr. Microbiol. Appl. Sci., 4(5): 872-881.

5. Yadav H., Gothwal R. K., Mathur S. and Ghosh P. (2015). Bioactivation of Jhamarkotra rock phosphate by a thermotolerant phosphate solubilizing bacterium Bacillus sp. BISR-HY63 isolated from phosphate mines. Arch. Agron. Soil Sci., 61 (8): 1125-1135. DOI: 10.1080/03650340.2014.980239.

6. Yadav H., Gothwal R. K., Solanki P. S., Nehra S, Sinha-Roy S. and Ghosh P. (2015). Isolation and characterization of thermo-tolerant phosphate-solubilizing bacteria from a phosphate mine and their rock phosphate solubilizing abilities. Geomicrobiology J., 32 (6): 475-481. DOI: 10.1080/01490451.2014.943856.

7. Meena S., Gothwal R. K., Mohan M. K. and Ghosh P. (2014). Production and purification of a hyperthermostable chitinase from Brevibacillus formosus BISR-1 isolated from the Great Indian Desert soils. Extremophiles, 18 (2): 451-462.

8. Meena S., Gothwal R. K., Saxena J., Mohan M. K. and Ghosh P. (2014). Chitinase production by a newly isolated thermotolerant Paenibacillus sp. BISR-047. Ann. Microbiol., 64 (2): 787-797. DOI 10.1007/s13213-013-0715-9.

9. Yadav H., Gothwal R. K., Nigam V. K., Sinha-Roy S. and Ghosh P. (2013). Optimization of culture conditions for phosphate solubilization by a thermo-tolerant phosphate-solubilizing bacterium Brevibacillus sp. BISR-HY65 isolated from phosphate mines. Bioc. Agricul. Biotechnol., 2 (3): 217-225.

10. Nehra S., Gothwal R. K., Chandra S., Meena P., Mohan M. K., Trivedi P. C. and Ghosh P. (2012). Screening and characterization of Pseudomonas fluorescence: A biological control against wilt fungus in Cajanus cajan (L.) Millsp. LS–An International Journal of Life Sciences, 1 (3): 175-185.

11. Rajvanshi A., Kulshreshtha S., Nigam V. K., Gothwal R. K. and Pareek R. P. (2011). Antagonistic properties of different bacteria isolated from salads. Asian J. Exp. Sci., 25 (2): 45-51.

12. Gothwal R. K., Mohan M. K. and Ghosh P. (2010). Synthesis of low cost adhesives from pulp and paper industry waste. J. Sci. Ind. Res., 69: 390-395.

13. Nigam, V. K., Khandelwal, A. K., Gothwal R. K., Mohan M. K., Choudhury, B., Vidyarthi, A. S. and Ghosh P. (2009). Nitrilase-catalysed conversion of acrylonitrile by free and immobilized cells of Streptomyces sp. J. Biosciences, 34 (1): 21-26.

14. Gothwal R. K., Nigam V. K., Mohan M. K., Sasmal D. and Ghosh P. (2008). Screening of nitrogen fixers from rhizospheric bacterial isolates associated with important desert plants. Appl. Ecol. Environ. Res., 6 (2): 101-109.

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18. Choudhury B., Aggarwal P., Gothwal R. K., Mantri R., Mohan M. K. and Ghosh P. (2006). Biobleaching of nonwoody pulps using xylanase of Bacillus brevis BISR-062. Appl. Biochem. Biotechnol., 128 (2): 159-170.

19. Gothwal R. K., Nigam V. K., Mohan M. K., Sasmal D. and Ghosh P. (2006). Small scale DNA extraction from soils of hot arid zones. http://www.protocol-online.org.

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Accession Numbers: KC534876, KC534875, KC534874, KC534873, JX195721, JX195720, JX195722, JX141618, JX049588, KC750085, KC847051, KF537620, KR866375.

REVIEW ARTICLES

1. Raj Kumar Gothwal, Snehanshu Shekhar, Vishwambhar Pathak, Pooran Singh Solanki, Alok Kumar Varshney and Sampat Nehra (2026). Bio-Cyber-Physical Systems (Bio-CPS): Concepts and Applications. In: Intelligent Cyber-Physical Systems: Integrating AI, ML, IoT, and Data Analytics for Advanced Electrical and Mechanical Engineering Applications", Editors Vivek Kumar, Jai Gopal Gupta, Gajendra Singh and Padmini Singh. Springer (Accepted).

2. Nupur Chhapdia, Pooja Sharma, Yuttasha, Monojit Majhi, A. Karpaga Kalyani, Dipali Bhise, Erica Zinni Nehra, Alok Kumar Varshney, Pooran Singh Solanki, Sampat Nehra, and Raj Kumar Gothwal (2025). Impact of Soil Microbiome on Climate Change. In: Chemo-Signals in the Rhizomicrobiome", edited by ; Jain, S., Vaishnav, A., Sharma, D. Sidana, A. and Choudhary, D.K. CRC Press (Taylor & Francis Group) (Accepted).

3. Monojit Majhi, Nupur Chhapdia, Yuttasha, Pooja Sharma, A. Karpaga Kalyani, Dipali Bhise, Erica Zinni Nehra, Alok Kumar Varshney, Pooran Singh Solanki, Sampat Nehra, and Raj Kumar Gothwal (2025). Chemo-Signal in Soil Microorganisms–Plant Interaction.In: Chemo-Signals in the Rhizomicrobiome", edited by ; Jain, S., Vaishnav, A., Sharma, D. Sidana, A. and Choudhary, D.K. CRC Press (Taylor & Francis Group) (Accepted).

4. Pooja Sharma, Yuttasha, Nupur Chhapdia, Monojit Majhi, Erica Zinni Nehra, Alok Kumar Varshney, Pooran Singh Solanki, Sampat Nehra, and Raj Kumar Gothwal (2025). Rhizomicrobiome Mediated Amelioration of Drought Stress. In: Chemo-Signals in the Rhizomicrobiome", edited by ; Jain, S., Vaishnav, A., Sharma, D. Sidana, A. and Choudhary, D.K. CRC Press (Taylor & Francis Group) (Accepted).

5. Yuttasha, Pooja Sharma, Nupur Chhapdia, Monojit Majhi, Erica Zinni Nehra, Alok Kumar Varshney, Pooran Singh Solanki, Sampat Nehra, and Raj Kumar Gothwal (2025). The Influence of Chemosignals on Microbe–Microbe Interactions.In: Chemo-Signals in the Rhizomicrobiome", edited by ; Jain, S., Vaishnav, A., Sharma, D. Sidana, A. and Choudhary, D.K. CRC Press (Taylor & Francis Group) (Accepted).

6. Varshini G, Aakriti, Jha, Erica Zinnia Nehra, Sampat Nehra, and Raj Kumar Gothwal (2025). Case Studies and Practical Applications in Parasitology. In:Parasitology: From Host Interactions to Disease Management, Edited by Arti Gupta & Nagendra Singh, Springer Nature (Switzerland)(Accepted).

7. Aakriti Jha, Varshini G, Erica Zinnia Nehra,Sampat Nehra, and Raj Kumar Gothwal (2025). Systematics in Parasitology. In: Parasitology: From Host Interactions to Disease Management, Edited by Arti Gupta & Nagendra Singh, Springer Nature (Switzerland)(Accepted).

8. Yuttashaa, Pooja Sharma, Nupur Chhapdia, Monojit Majhi, Erica Zinnia Nehra, Sampat Nehra, Raj Kumar Gothwal (2025). Molecular Evolution of Genes and Advancements in Molecular Techniques for Disease Resistance. In: Molecular Breeding for Disease Resistance: Recent Advances. Bentham Sciences.(Accepted)

9. Aruna Shekhar Nannivala Chandra Shekhar, Prachiben P. Patel, Jessica Jeejan, Erica Zinnia Nehra, Sampat Nehra, Raj Kumar Gothwal (2025). Mapping-by-Sequencing and Marker-Assisted Selection (MAS) for Disease Resistance. In: Molecular Breeding for Disease Resistance: Recent Advances. Edited by Neeraj Verma, AP Pandey & ALN Rao.Bentham Sciences.(Accepted)

10. Aarushi Sachdeva, Erica Zinnia Nehra, Raj Kumar Gothwal, Sampat Nehra (2025). Intellectual Property Rights for Genetically Modified Crops. In: Molecular Breeding for Disease Resistance: Recent Advances. Edited by Neeraj Verma, AP Pandey & ALN Rao.Bentham Sciences.(Accepted)

11. Pooja Sharma, Yuttasha, Nupur Chhapdia, Monojit Majhi, Erica Zinnia Nehra, Sampat Nehra, Raj Kumar Gothwal (2025). Climate Change and Disease Resistance: Molecular Breeding Techniques to Develop Resilient Varieties Against New and Evolving Threats. In: Molecular Breeding for Disease Resistance: Recent Advances. Edited by Neeraj Verma, AP Pandey & ALN Rao.Bentham Sciences.(Accepted)

12. Jessica Jeejan, Aruna Shekhar Nannivala Chandra Shekhar, Prachiben P. Patel, Erica Zinnia Nehra, Raj Kumar Gothwal, Sampat Nehra (2025). Gene Pyramiding, Genome Selection (GS), Fine Mapping, Allele Mining Strategies, Pathways For Disease Resistance. In: Molecular Breeding for Disease Resistance: Recent Advances. Edited by Neeraj Verma, AP Pandey & ALN Rao. Bentham Sciences. (Accepted)

13. Prachiben P. Patel, Jessica Jeejan,Aruna Shekhar Nannivala Chandra Shekhar, Erica Zinnia Nehra, Sampat Nehra, Raj Kumar Gothwal (2025). Transformation for Disease Resistance: Recent Approaches. In: Molecular Breeding for Disease Resistance: Recent Advances.Edited by Neeraj Verma, AP Pandey & ALN Rao. Bentham Sciences. (Accepted).

14. Gothwal R K, Nehra S, Sinha, P, Shekhar NC A and Ghosh P. (2024). Proteomic-driven approaches for unraveling plant-microbe interactions. In: Plant Proteomics: implications in growth, quality improvement and stress resilience. A R Choudhary (Ed.), CRC Press (Taylor & Francis Group), p. 114-153.ISBN 9781032395920, DOI: 10.1201/b23255-10.

15. Gothwal R K, Nehra S, Varshney A K, Solanki P S and Ghosh P (2024). Endophytes: distribution, molecular characterization and biodiversity evaluation. In: Industrial Applications of Soil Microbes, Volume-4, Bentham Science Publisher. P. 144-160. DOI: 10.2174/9789815124996124040012.

16. Nehra S, Gothwal R K, Varshney A K, Solanki P S, Meena P, Trivedi P C and Ghosh P (2023). Alleviation of salinity stress by microbes. In: Industrial Applications of Soil Microbes, Volume-2, p. 145-174, Bentham Science Publisher. DOI: 10.2174/9789815050264123020012.

17. Nehra S, Gothwal R K, Varshney A K, Meena P, Trivedi P C and Ghosh P (2024). Carotenoids in microorganisms and their applications. In: Industrial Application of Soil Microbes, Volume-3, p. 228-239,

Bentham Science Publisher.Print ISSN 2811-0765, Online ISSN2811-0773, DOI: https://doi.org/10.2174/9789815079753124030O.

18. Nehra S, Gothwal R K, Varshney A K, Meena P, Ghosh P and Trivedi P C (2022). Bioactives and Pharmacology of Capparis decidua (Forssk.) Edgew. (Syn.: Capparis aphylla Roth) (Family: Capparaceae)". In: Bioactives and Pharmacology of Medicinal plants. Volume-1. (Ed.) T. Pullaiah, Apple Academic - CRC Press, Taylor and Francis Group. p. 207-220. ISBN: 9781774910283. https://doi.org/10.1201/9781003281658.

19. Nehra S, Gothwal R K, Dhingra S, Varshney A K, Solanki P S, Meena P, Ghosh P and Trivedi P C (2022). Mechanism of Antagonism: Hyperparasitism and Antibiosis. In: Microbial Biocontrol: Sustainable Agriculture and Phytopathogen Management. Volume 1. Ajay Kumar (Ed.), Springer Nature Switzerland AG. ISBN: 9783030875114 pp. 257-277.

20. Nehra S, Gothwal R K, Varshney A K, Solanki P S, Chandra S, Meena P, Trivedi P C and Ghosh P (2021). Biomanagement of Fusarium spp. associated with pulse crops. In: Food Security and Plant Disease Management. (Eds.) A Kumar and S Droby, Woodhead Publishing, Elsevier, pp. 423-452. ISBN: 9780128218433.

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22. Nehra S, Gothwal R K, Varshney A K, Solanki P S, Chandra S, Meena P, Trivedi P C and Ghosh P (2021). Bio-management of Fusarium spp. associated with fruit crops. (Eds.) V K Sharma, M P Shah, S Parmar and A Kumar. In: Fungi Bio-Prospects in Sustainable Agriculture, Environment and Nano-technology, Volume-1, pp 475-505. Academic Press, Elsevier, ISBN: 9780128213940.

23. Nehra S, Gothwal R K and Ghosh P (2014). Biodiversity conservation: An Indian agenda for the millennium. In: Biodiversity in India: Assessment, Scope and Conservation, Lambert Academic Publishing (LAP), Germany, pp. 323-340.

24. Nehra S, Gothwal R K, Chandra S, Meena P, Mohan M K and Ghosh P (2014). Role of microbes in bioremediation. In: Applied Microbiology (Ed.) Nehra S, Pointer Publication, Jaipur, pp. 232-248.

25. Nehra S, Gothwal R K, Chandra S Meena P, Mohan M K and Ghosh, P (2014). Rhizobium: Nitrogen fixation and agriculture, In: Biofertilizers for sustainable agriculture (Ed.) Nehra S., Pointer Publication, Jaipur, pp. 209-221.

26. Karkho R, Gothwal R K and Nehra S (2014). The treatment of leprosy using various drugs and antileprotic plants: A Review. In: Plant Disease Management and Microbes (Ed.) Nehra S., Aavishkar Publication, Jaipur, pp. 250-258.

27. Saxena T, Varshney A K, Gothwal R K, Kaushik P, Mohan M K and Nehra S (2014). Immunological Techniques-Role in disease management. In: Plant Disease Management and Microbes (Ed.) Nehra S., Aavishkar Publication, Jaipur, pp. 259-268.

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31. Gothwal R K, Nigam V K, Mohan M K and Ghosh P (2011). Rhizosphere associated plant-microbial interactions in Thar arid ecosystem. In: Plant Tissue Culture and Applied Plant Biotechnology. (Eds.) Ashwani Kumar and Shikha Roy, Avishkar Publishers, Distributors, India, pp. 292-313.

PRESENTATIONS IN CONFERENCES/WORKSHOPS/SYMPOSIA

1. Pallawi Singh, Garima Rani, RK Gothwal, Sampat Nehra and PS Solanki (2026). In-Silico approach for Exploring Antimicrobial Coating Challenges for Orthodontic Implants. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

2. Rahul Sangwan, RK Gothwal, Sampat Nehra and PS Solanki (2026). Transcriptomic Insights into the Antifungal Drug Response in Cryptococcus neoformans: Metabolomics Approach. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

3. Chitrita Nag, A.K. Kalyani, Nupur Chhapdia, P. S. Solanki, R. K. Gothwal & S. Nehra (2026). Logic-Gated CAR-T cell system MUC1 and HER2 in MCF-7 cell lines for breast cancer. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

4. Gothwal R. K. (2026). Desert Microbes: A New Frontier for Sustainable Agriculture. Invited talk at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

5. Anshu Sheoran, Sampat Nehra, RK Gothwal and PS Solanki (2026). Atomic interactions and their role in protein structural stability using Bioinformatics approaches. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

6. Drishti Samadhiya, Sampat Nehra, RK Gothwal and PS Solanki (2026). A Bioinformatics-Driven Framework for the Systematic Analysis and Interpretation of Microbial Biochemical Profiles. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

7. A.K. Kalyani, Nupur Chhapdia, A. K. Varshney, P. S. Solanki, R. K. Gothwal & S. Nehra (2026). Investigating effect of Metformin on MCF-7 Breast Cancer Cell Viability. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development"

organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

8. Manvi Katiyar, Sampat Nehra, RK Gothwal and PS Solanki (2026). Computational Identification of Immunogenic Epitopes from DENV (E) Envelope Protein for Developing a Vaccine Lead. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

9. Monojit Majhi, Yuttasha, Deepali Bhise, S. Nehra, A. K. Varshney, P. S. Solanki & R. K. Gothwal (2026). Prevalence of Diarrhea-Associated Enteric Bacterial Pathogens in Children Aged Between Years 1-5. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

10. Neha Ganeriwala, R. K. Gothwal, Sampat Nehra and PS Solanki (2026). Immunoinformatics-Based Multi-Epitope Vaccine Candidate for Helicobacter pylori. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

11. Nupur Chhapdia, A Kalyani, R K Gothwal, P S Solanki and Sampat Nehra (2026). Fenugreek-Based Nanoparticles for Targeted Breast Cancer Therapy. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

12. Pooja Sharma, S. Nehra, A. K. Varshney, P. S. Solank & R. K. Gothwal (2026). Investigation of Rhizobacterial Communities in native Poaceae members along with a Natural Aridity Gradient in Rajasthan. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

13. Vishishtaa Pandit, Vipin Ranga, Sampat Nehra, RK Gothwal, P.S. Solanki (2026). Linking Cellular miRNA Dysregulation to Women's Health Outcomes: A Comparative Study of Ovarian and Cervical Cancers. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

14. Yuttasha, Monojit Majhi, S. Nehra, A. K. Varshney, P. S. Solank & R. K. Gothwal (2026). Prevalence of Salmonella and Shigella spp. causing Paediatric Diarrhea. Research poster presented at International Conference on "Cells to Communities: Interlinking Life Sciences, Forestry, and Social Development" organized by Department of Botany, IIS (deemed to be University), Jaipur, India during on February 5-6, 2026.

15. Ankush Sharma, Pallawi Singh, Rahul Sangwan, A. K. Varshney, S. Nehra, P. S. Solanki and R. K. Gothwal (2025). Phytochemical Targeting of MRSA: An In Silico Approach. Research poster presented at National Conference on Bio 3: Biotechnology, Biomanufacturing and Bioeconomy organized by Thapar Institute of Engineering & Technology, Patiala, Punjab during March 20-22, 2025.

16. Yuttasha, Pooja Sharma, A. K. Varshney, S. Nehra, P. S. Solanki and R. K. Gothwal (2025). Understanding The Role of Bacterial Pathogens and Environmental Factors in the Incidence of Pediatric Diarrhea.

Research poster presented at National Conference on Bio 3: Biotechnology, Biomanufacturing and Bioeconomy organized by Thapar Institute of Engineering & Technology, Patiala, Punjab during March 20-22, 2025.

17. Pallawi Singh, Ankush Sharma, Rahul Sangwan, A. K. Varshney, S. Nehra, P. S. Solanki and R. K. Gothwal (2025). Unveiling the Immunological Link Between Autoimmune Diseases and Alzheimer's Disease. Research poster presented at National Conference on Bio 3: Biotechnology, Biomanufacturing and Bioeconomy organized by Thapar Institute of Engineering & Technology, Patiala, Punjab during March 20-22, 2025.

18. Pooja Sharma, Yuttasha, A. K. Varshney, S. Nehra, P. S. Solanki and R. K. Gothwal (2025). Exploring The Plant Growth Promoting Rhizobacterial Communities Associated With Native Members of Poaceae Across a Natural Aridity Gradient. Research poster presented at National Conference on Bio 3: Biotechnology, Biomanufacturing and Bioeconomy organized by Thapar Institute of Engineering & Technology, Patiala, Punjab during March 20-22, 2025.

19. Rahul Sangwan, Pallawi Singh, Ankush Sharma, A. K. Varshney, S. Nehra, P. S. Solanki and R. K. Gothwal (2025). Environmental Factors and Gene Expression in Non-Smoker Lung Cancer: A Bio-informatics Analysis. Research poster presented at National Conference on Bio 3: Biotechnology, Biomanufacturing and Bioeconomy organized by Thapar Institute of Engineering & Technology, Patiala, Punjab during March 20-22, 2025.

20. Pooja Sharma, N. Chhapdia, A. K. Varshney, P. S. Solanki, S. Nehra & R. K. Gothwal (2025). The metagenomic landscape of rhizobacterial communities in a natural aridity gradient. Research paper presented at BIT Research Colloquium organized by BIT Mesra, during July 13-14, 2025.

21. Yuttasha, M. Majhi, A.K Varshney, P.S. Solanki, S. Nehra and R. K. Gothwal (2025). Prevalence of bacterial pathogens causing paediatric diarrhea: An empirical culture-based analysis. Research paper presented at 5th BIT Research Colloquium organized by BIT Mesra, during July 13-14, 2025.

22. Manvi Katiyar, Sampat Nehra, R. K. Gothwal and P. S. Solanki (2025). In-silico designing of vaccine lead against Dengue virus using a multi-epitope based computational approach. Research paper presented at 5th BIT Research Colloquium organized by BIT Mesra, during July 13-14, 2025.

23. Poster presentation on "Database of molecular based identification of human fungal pathogens" in International Conference on Recent Trends in Engineering and Science (RTES–2023) organized on May 2-3, 2023. P. S. Solanki, R. K. Gothwal, Monisha Singhal, A. K. Varshney, S. Nehra, P. Tewari and P. Ghosh.

24. Pooja Sharma, Soniya Kanwar, Gaurang Sharma, Shivani Nagar, Ireena Bohra, Anusha Patel, Bhagyashri Karale, Sampat Nehra, R. K. Gothwal and P. Ghosh (2023). To study the complex hydrocarbon utilization potential of bacteria isolated from hot and arid regions of Rajasthan using turbidometry. Research paper presented at International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

25. Gaurang Sharma, Shivani Nagar, Ireena Bohra, Pooja Sharma, Soniya Kanwar, Anusha Patel, Bhagyashri Karale, Sampat Nehra, R. K. Gothwal and P. Ghosh (2023). Studies on short chain gaseous hydrocarbon utilizing bacteria isolated from the Thar Desert soils of Rajasthan. Research paper presented at International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

26. Bhagyashri Karale, R. K. Gothwal, Sampat Nehra, Sangeeta Sawant, A. K. Varshney, Anusha Patel, P. S. Solanki and P. Ghosh (2023). Interaction studies on viral genome-linked protein (VPg) of Plum pox virus with DEAD-box RNA helicase-like (PpDDXL) plant protein of Prunus persica. Research paper presented at

International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

27. Bhagyashri Karale, R. K. Gothwal, Sampat Nehra, A. K. Varshney, P. S. Solanki, Anusha Patel, M. P. Punia and P. Ghosh (2023). Development of small-scale community DNA extraction method for improved downstream processing from hot and arid region soils of Rajasthan to explore petroleum hydrocarbon utilizing bacterial diversity. Research paper presented at International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

28. Anusha Patel, Sampat Nehra, R. K. Gothwal, Sanjiva Bimal, A. K. Varshney, P. S. Solanki, Bhagyashri Karale and P. Ghosh (2023). A study on immunological cell interaction with Leishmania donovani antigen in the BALB/c mice to establish relation between malnourishment and globally affecting disease-Leishmaniasis.

29. Anusha Patel, R. K. Gothwal, Sampat Nehra, A. K. Varshney, P. S. Solanki, Bhagyashri Karale, M. P. Punia and P. Ghosh (2023). Dehydrogenase enzyme-based detection of petroleum hydrocarbon utilizing activity of bacteria using redox indicator dye in hot arid region soils of Rajasthan. Research paper presented at International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

30. Akash HC, R. K. Gothwal, Sampat Nehra, A. K. Varshney, P. S. Solanki, Anusha Patel, Bhavana AG, Bhagyashri Karale and P. Ghosh (2023). Diversity of microbial mediated petroleum hydrocarbon degrading genes in the environment. Research paper presented at International Conference on Contemporary Trends in Multidisciplinary Research & Innovation (ICCTMRI) organized by BIT Mesra, Jaipur-Campus during July 27-28, 2023.

31. Vidhi choudhary, Sampat Nehra and Raj Kumar Gothwal 2022. A Novel, Sustainable and Economical Way to Generate Electrical Power: Converting Potential Energy of the Moving Vehicle into Electric Energy on the Roads. In: Technorazz 2022, 30/9/2022-1/10/2022, presentation for Young Researcher and Innovation Award, JNU, Jaipur.

32. Gothwal R. K., Karale B., Joseph A., Varshney A.K., Nehra S., Solanki P. S., Punia M. P. and Ghosh P. (2022). An efficient metagenomic DNA extraction method from Thar Desert soils for exploring petrolium hydrocarbon utilizing bacterial diversity. Presented in 4th Biological Engineering Society Conference BESCON-2022 'Smart and Sustainable Bioprocessing through gene and process scale collaboration' organized by IIT Kharagpur at Bose Institute, Kolkata during November 4-6, 2022.

33. Solaki P. S., Gothwal R. K., Varshney A. K., Nehra S. and Ghosh P. (2022). Database of biochemical tests of pathogenic Enterobacteriaceae family. Presented in 4th Biological Engineering Society Conference BESCON-2022 'Smart and Sustainable Bioprocessing through gene and process scale collaboration' organized by IIT Kharagpur at Bose Institute, Kolkata during November 4-6, 2022.

34. Nehra S., Patel A., Varshney A. K., Solanki P. S., Punia M. P., Gothwal R. K. and Ghosh P. (2022). Development of a redox dye based novel and reliable method for detecting hydrocarbon utilizing bacteria directly from the soil samples. Presented in 4th Biological Engineering Society Conference BESCON-2022 'Smart and Sustainable Bioprocessing through gene and process scale collaboration' organized by IIT Kharagpur at Bose Institute, Kolkata during November 4-6, 2022.

35. Meena P., Gothwal R. K. and Nehra S. (2021). Biodegradation of textile azo dyes by bacteria isolated from dyeing industry effluent. Presented in 44th All India Botanical Conference of the IBS and National

Symposium on Plant Science Research in Present Scenario: Opportunities and Challenges, organized by JNVU, Jodhpur during October 18-20.

36. Pooran Singh Solanki, Raj Kumar Gothwal, Krishna Mohan Medicherla (2019). PIASP: A tool to determine structurally conserved atomic interactions in protein-ligand Complexes. Presented in Inbix'19: Indian Conference on Bioinformatics organized by Hansraj Mahila Maha Vidyalay (HMV), Jalandhar, India, February 22-23.

37. Gothwal R. K. (2000). In vitro and biochemical studies in Chlorophytum borivilianum. National Symposium on Prospects and Potentials of Plant Biotechnology in India in the 21st Century and 23rd Annual Meeting of PTCA, India organized by Jai Narayan Vyas University, Jodhpur, Rajasthan during October 18-21.

38. PS Solanki, Monisha Singhal, Pragya Chaturvedi, Priyanka Yadav, Gothwal R. K., M K Mohan (2018). Database of specific primers and probes of water pathogens. Presented in International Conference on Bioinformatics and Computational Biology (INCOB-2018) organized by JNU, New Delhi, September 26-28.

39. Solanki P. S., Gothwal R. K., Mohan M. K. and Ghosh P. (2018). Poster presentation at XXIX Annual Bioinformatics Review cum Coordinators Meeting organized by MS University, Tirunelveli during February 3-4, 2018.

40. Meena S., Gothwal R. K., Jyoti Saxena, A. K. Varshney, P. S. Solanki, S. Nehra, Mohan M. K. and Ghosh P. (2017). Utilization of sea food waste for chitinase production by Brevibacillus formosus BISR-1 and Paenibacillus sp. BISR-047 in National Conference of Biological Engineering Society on Biological Engineering in 21st Century organized by NSIT, New Delhi during September 8-9, 2017.

41. Dadheech P. K. and Gothwal R. K. (2016). Significance of DNA-barcoding in the systematics of fungi: An overview. National conference on Fungal Biotechnology and 43rd Annual Meeting of the Mycological Society of India organized by Birla Institute of Scientific Research, Jaipur during November 16-18, 2022.

42. Gothwal R. K., Meena S., Sharma Y. K., Nehra S., Solanki P. S., Mohan M.K. and Ghosh P. (2014). A hyper-thermostable chitinase from Paenibacillus sp. BISR-047: Its characterization and biocontrol potential. 55th Annual Conference of AMI and National Conference on Empowering Mankind with Microbial Technologies organized by TNAU, Coimbatore, India during November 12-14 (AM56).

43. Yadav H., Gothwal R. K., Nehra S., Solanki P. S. and Ghosh P. (2014). Isolation and characterization of thermotolerant bacterial strains from Jhamarkotra phosphate mines producing alkaline phosphatase. 55th Annual Conference of AMI and National Conference on Empowering Mankind with Microbial Technologies organized by TNAU, Coimbatore, India during November 12-14 (AM74).

44. Gothwal R. K. (2013). Mineral-Microbe interactions: a driver for mineral weathering. Contact programme workshop on Earth Surface Processes organized by IIT Gandhinagar during December 9-10.

45. Nehra S., Gothwal R. K., Mohan M. K. and Ghosh P. (2013). Characterization of rhizobacteria for plant growth promoting activities and to manage wilt disease complex in chilli (Capsicum annuum L.). Emerging Trends in Nanobiotechnology organized by Jaipur National University, Jaipur during November 29-30.

46. Nehra S., Gothwal R. K., Solanki P., Mohan M. K. and Ghosh P. (2013). Genotyping of antifungal compounds producing plant growth promoting rhizobacteria Pseudomonas fluorescence. Emerging Trends in Nanobiotechnology organized by Jaipur National University, Jaipur during November 29-30.

47. Gothwal R. K., Meena S., Nehra S., Solanki P. S., Mohan M. K. and Ghosh P. (2013). Chitinase production and biocontrol potential of a newly isolated chitinolytic bacterium Brevibacillus formosus BISR-1. 54th Annual Conference of AMI and International Symposium on Frontier Discoveries and Innovations in

Microbiology and its Interdisciplinary Relevance organized by MDU, Rohtak, India during November 17-20.

48. Yadav H., Sharma S., Gothwal R. K., Nigam V. K., Mohan M. K., Sinha-Roy S. and Ghosh P. (2012). Rock phosphate solubilization by thermo-tolerant bacterial strains isolated from Jhamarkotra phosphate mines of Rajasthan. 53rd Annual Conference of AMI & International Conference on Microbial World: Recent Innovations and Future Trends organized by KIIT University, Bhubaneshwar, India during November, 22-25.

49. Meena S., Solanki P. S., Nehra S., Gothwal R. K., Mohan M. K. and Ghosh P. (2012). Optimization of culture conditions for chitinase production by Paenibacillus sp. BISR-047. 53rd Annual Conference of AMI & International Conference on Microbial World: Recent Innovations and Future Trends organized by KIIT University, Bhubaneshwar, India during November, 22-25.

50. Chitranshi K., Meena S., Sharma Y. K., Solanki P. S., Nehra S., Gothwal R. K., and Ghosh P. (2012). Antifungal activity of chitinolytic bacterial strains and their potential for biocontrol of important fungal phytopathogens of cumin. 53rd Annual Conference of AMI & International Conference on Microbial World: Recent Innovations and Future Trends organized by KIIT University, Bhubaneshwar, India during November, 22-25.

51. Meena S., Gothwal R. K., Sharma Y. K. and Ghosh P. (2012). Production of chitinase by Paenibacillus sp. BISR-047 and its potential for the biocontrol of important fungal phytopathogens of cumin. 4th Indian Youth Science Congress on Young Scientists and Shaping India's Future at Nagpur, Maharashtra during November 9-11.

52. Gothwal R. K. (2012). Chitinolytic enzymes. National Seminar on Science for Shaping the Future of India organized by CCT, University of Rajasthan, Jaipur during October 6-7.

53. Solanki P. S., Gothwal R. K., Mohan M. K., Ghosh P. and Kothari S. L. (2012). Comparative analysis of secondary structure elements (SSEs) in extremophiles using CASSP. National Symposium on Bioinformatics: Challenges in the post-genomic era organized by University of Jammu during February 2.

54. Pareek R. P., Nigam V. K., Mohan M. K., Dey S., Gothwal R. K. and Ghosh P. (2011). Development of liquid biofertilizer formulation for Rhizobium in Cicer arietinum. International Symposium on Plant Biotechnology towards Tolerance to Stresses and Enhancing Crop Yield organized by BIT, Mesra, Ranchi, Jharkhand during September 28 - October 1, 2011.

55. Meena S., Gothwal R. K., Nigam V. K., Mohan M. K. and Ghosh P. (2010). Studies on chitin solubilizing bacterial isolates from hot arid zones of Rajasthan. International Symposium on Recent Advances in Cross-disciplinary Microbiology: Avenues & Challenges organized by BIT, Mesra, Ranchi during December 14-17.

56. Sinash S., Yadav H., Gothwal R. K., Nigam V. K., Mohan M. K., Sinha-Roy S. and Ghosh P. (2010). Phosphate solubilizing rhizobacteria from sandy soils of Jaipur region. International Symposium on Recent Advances in Cross-disciplinary Microbiology: Avenues & Challenges organized by BIT, Mesra, Ranchi during December 14-17.

57. Akanksha, Gothwal R. K., Nigam V. K., Mohan M. K. and Ghosh P. (2008). Studies on antifungal metabolites produced by endophytic and rhizobacterial isolates of Catharanthus roseus. National Symposium on Biotechnology in plant disease management for sustainable crop protection organized by Agharkar Research Institute, Pune during September, 17-18.

58. Nigam V. K., R. Verma, Gothwal R. K., S. Kundu and Ghosh P. (2005). Evaluation of cephalosporin-C production at different inoculum conditions. National conference on Bioprocess Engineering and Biotechnology: Trends and Opportunities organized by BIT, Mesra, Ranchi during March 17-19.

59. Gothwal R. K., Nigam V. K., Mohan M. K., Sasmal D. and Ghosh P. (2005). Rhizospheric community analysis of desert plants: An approach for optimization of DNA amplification. National conference on Bioprocess Engineering and Biotechnology: Trends and Opportunities organized by BIT, Mesra, Ranchi during March 17-19.

60. Dadheech P. K., Gothwal R. K., Sinha S. and Mohan M. K. (2004). Cyanobacterial diversity in the Great Indian Desert inferred with polyphasic approach. 16th Symposium of the International Association for Cyanophyte Research at Luxembourg, Germany during August 30 - September 3.

61. Gothwal R. K. (2004). Isolation of bacterial genomic DNA, its electrophoresis and PCR amplification. Science Congress organized by Navodaya Vidyalaya, Paota, Jaipur, Rajasthan during November.

62. Gothwal R. K. (2004). Use of lignin for wood adhesive. National Seminar on Biomass as Source of Energy and Chemicals organized by Department of Chemical Engineering & Technology, BHU, Varanasi during February 28-29.

ORGANIZATION OF 6-MONTHS CERTIFICATE COURSES

1. Co-coordinator of the 6-Months Certificate course (1-Semester) on Computer Assisted Drug Design (CADD) during September 3, 2021 to March 2022, jointly organized by Birla Institute of Scientific Research and Design and Innovation Centre (Online Mode).

2. Co-coordinator of the 6-Months Certificate course (1-Semester) on Computer Assisted Drug Design (CADD) during September 3, 2022 to March 2023, jointly organized by Birla Institute of Scientific Research and Design and Innovation Centre (Online Mode).

3. Co-coordinator of the 6-Months Certificate course (1-Semester) on Computer Assisted Drug Design (CADD) during September 15, 2023 to March 2, 2024, jointly organized by Birla Institute of Scientific Research and Design and Innovation Centre, University of Rajasthan, Jaipur (Online Mode).

4. Co-coordinator of the 6-Months Certificate course (1-Semester) on Computer Assisted Drug Design (CADD) during September 27, 2024 to March 15, 2025, jointly organized by Birla Institute of Scientific Research and Design and Innovation Centre, University of Rajasthan, Jaipur (Online Mode).

5. Co-coordinator of the 6-Months Certificate course (1-Semester) on Computer Assisted Drug Design (CADD) during September 26, 2025 to March 7, 2026 jointly organized by Birla Institute of Scientific Research and Design and Innovation Centre, University of Rajasthan, Jaipur (Online Mode).

ORGANIZATION OF TRAINING PROGRAMMES

1. Co-coordinator of the 6-Months dissertation programme being conducted every year twice in the months of January-June and July-December.

2. Co-coordinator of the 6-Weeks Summer Training Programme in Microbiology; Molecular Biology; Fermentation, Protein Extraction and Estimation; Bioinformatics, drug design and next generation sequencing; Laboratory equipment and Bioinformatics software for data analysis being conducted every year in the months of May-July.

ORGANIZATION OF CONFERENCES/SEMINARS/WORKSHOPS

1. Coordinator, an Interactive Session on "Current Research Goals" organized by the Department of Biotechnology, Birla Institute of Scientific Research, Jaipur under the aegis of the National Initiatives for Design Innovation (NIDI), Department of Higher Education (MoE), Government of India held on March 27, 2026.

2. Convener, One-Day Workshop on "Mammalian Cell Culture: Theory to Practical" organized by the Department of Biotechnology, Birla Institute of Scientific Research, Jaipur during March 19, 2026.

3. Convener, DST sponsored International Conference on "Current Trends in Multidisciplinary Domains of Research (ICCTMDR-2026)" under the aegis of National Initiatives for Design Innovation (NIDI), MoE, GoI organized by the Department of Biotechnology, BISR, Jaipur during March 13-14, 2026.

4. Session Coordinator, "DNA Extraction Hands-on Activity" in Rajasthan Vigyan Mahotsav – National Science Day 2026 organized by the DST, GoR in association with Vigyan Bharti Rajasthan and MNIT, Jaipur at VLTC Complex, Ramanujan Hall, MNIT, Jaipur during February 26-28, 2026.

5. Coordinator, Indian National Academy of Engineering (INAE) Annual Convention at BISR, Jaipur held during December 12-14, 2019.

6. Co-Coordinator, 3-days National Workshop on Bioinformatics Applications in Modeling and Drug Design at BISR, Jaipur held during Feb 15-17, 2018.

7. Coordinator, Indian National Academy of Engineering (INAE) Youth Conclave at BISR, Jaipur held during August 11-12, 2017.

8. Joint Organizing Secretary, National Conference on Fungal Biotechnology and 43rd Annual Meeting of the Mycological Society of India held during November 16-18, 2016.

9. Coordinator, Indian National Academy of Engineering (INAE) Annual Convention held during 11-13 December 2014.

10. Co-Coordinator, XVI Annual BTISnet Coordinators Meet at BISR, Jaipur held, February 2005.

11. Co-Coordinator, 74th Annual Session of National Academy of Sciences, India at BISR and UOR, Jaipur, December 2004.

EXTRACURRICULAR ACTIVITIES

- National West-Zone Youth Festival organized by Devi Ahilya Vishwavidyalaya, Indore, Madhya Pradesh during 1988-89.
- Inter-University Cricket Tournament organized at Bharuch, Gujarat during 1990-91.
- Inter-University Tennis Tournament organized at Coimbatore, Tamil Nadu during 1989-90.

MEMBER OF PROFESSIONAL BODIES

- Founder Member, Biological Engineering Society (BESI)
- Honorary Life Member, Asian PGPR Society of Sustainable Agriculture
- Life Member, Bioinformatics Club for Experimenting Scientists (BIOCLUES) Society
- Contributing Member, American Society for Microbiology (ASM)

- Life Member, Association of Microbiologists of India (AMI)
- Life Member, Indian Science Congress Association (ISCA)
- Life Member, Asian Agri-History Foundation (AAHF)
- Life Member, Vigyan Bharti Rajasthan (VBR)
- Life Member, Mycological Society of India (MSI)
- Annual Member, Amateur Astronomers Society (AAS)
- Annual Member, Indian Phytopathological Society (IPS)

CURRENT SPONSORED PROJECTS

1. Studies Magnified Upon Variable/Multiple Causative Pathogens Culminating into Diarrheal Diseases in Children Aged Between Years 1- 5 (Department of Science and Technology Funded Project - Ongoing).

2. Impact of metformin on cancer cell viability and growth inhibition: a mechanistic investigation (Department of Science and Technology Funded Project - Ongoing).

3. Production of laccase enzyme by Pycnoporus sanguineus using low cost agri residues (Department of Science and Technology Funded Project - Ongoing).

4. A database of human pathogens related to food and waterborne diseases (Department of Science and Technology Funded Project - Ongoing).

5. Design Innovation Centra (DIC), One of the spoke to the Hub DIC, Centre for Converging Technologies (CCT), UoR, with a total grant of of Rs. 1.0 Crore. Department of Higher Education, Ministry of Education, Govt. of India funded Project - Ongoing.

6. GIS mapping of microbial population for their geographical position, geological and geochemical habitats to explore oil and gas reservoirs in Jaisalmer sub-basin of Thar Desert (NGP, Department of Science and Technology, Govt. of India Funded Project - Completed).

7. Bacterial profiling of Bisalpur drinking water distribution system: A comparative study from the source to the tap (Department of Science and Technology, Govt. of Rajasthan Funded Project – Completed).

8. Increased guar gum production, disease suppression and plant development by treatment of rhizobacteria in cluster bean (Department of Science and Technology, Govt. of Rajasthan Funded Project – Completed).

9. Rapid multiplex PCR based detection of water borne human bacterial pathogens (Department of Science and Technology, Govt. of Rajasthan Funded Project– Completed).

10. Multiparametric study of soil-microbe-phosphate system for optimizing the role of phosphate-solubilizing bacteria in enhancing the efficiency of rock phosphate as fertilizer in direct application (Department of Biotechnology, Govt. of India Funded Project– Completed).

TEXT AND REFERENCE BOOKS

1. Nehra, R. K. Gothwal and P. Ghosh (2014). Biodiversity in India: Assessment, Scope and Conservation, Lambert Academic Publishing (LAP), Germany. ISBN: 978-3659505508.

Dr. Ravindra Kumar

Joined Institute on: 1-Sep-2008

Assistant Professor

Physics

Ph.D. M.Tech., M.Sc. (Physics)

CONTACT ADDRESS

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Phone Residence: 9887936418

Email: ravindrakumar@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 17 Years

Research: 18 Years

PROFESSIONAL PROFILES

LinkedIn Profile: www.linkedin.com/in/ravindra-dixit-a4a653154

Google Scholar: yOECuTgAAAAJ

Web of Science: www.webofscience.com/wos/author/record/KXR-9361-2024

PROFESSIONAL QUALIFICATIONS

Ph.D. (Plasma Physics) from VGU, Jaipur in 2019.

M.Tech. in New Materials and Processing Technology from Birla Institute of Technology, Mesra Ranchi in 2007.

M.Sc. (Physics) from M.M. (PG) College Modinagar (CCS University, Meerut) in 2004.

B.Sc. (PCM) from S.S.V. (PG) College Hapur (CCS University, Meerut) in 2002.

JUNIOR RESEARCH FELLOW (JRF)

Junior Research Fellow in a Department of Science and Technology (DST) sponsered project in the Department of Physics BIT Mesra, Jaipur Campus (August, 2007 to August, 2008).

SUBJECTS TAUGHT AT BIT MESRA, JAIPUR CAMPUS

- Physics (B.Tech (AI)-II Sem)
- Physics Lab (B.Tech (AI)-II Sem)
- Electronic Devices
- Semiconductor Devices

- Engineering Electromagnetics
- Data Communication and Computer Networking
- Material Science
- NSS (B.Tech (AI)-IV Sem)
- Creative Art

ACADEMIC/ADMINISTRATIVE RESPONSIBILITIES AT BIT JAIPUR CAMPUS

- Member of Examination Cell
- Member of Doctoral Committee (DC)
- Member of NSS Team
- Member of SARC Team
- Faculty Co-mentor for Plasma Application Club

ORGANIZATION OF CONFERENCE/SYMPOSIA/WORKSHOP

1. Coordinator of International Workshop on Cold Plasma Technology and Applications, "CPTA-2025", at BIT Mesra, Jaipur Off-Campus, 6-8 February 2025.

2. Coordinator of IPR Outreach Program on Plasma, Its Applications and Nuclear Fusion, "PLASMA GYAAN-2022" (Training /Workshop for Students & Teachers on Basic Plasma Physics), 22-26 August 2022.

3. Coordinator of 36th National Symposium on Plasma Science and Technology (PLASMA-2021), 13-15 December 2021 at BIT Mesra, Jaipur Off-Campus.

WORKSHOPS/TRAINING/REFRESHER COURSE COMPLETED

1. One week FDP on NEP 2020 Orientation and Sensitization Programme under Malviya Mission TTP of UGC (online), 27 May – 04 July 2025.

2. One week FDP on Innovative Teaching & Learning Pedagogy organized by Research Foundation of India, 18-24 June 2025.

3. One week FDP on Universal Human Values (UHV-I) organized by AICTE (Online), 20-24 May, 2024.

4. Science Academies 68th Refresher Course in Experimental Physics during 10-25 March 2015 (16 Days) at Central University of Rajasthan, Ajmer.

5. Short Term Course on Structure and Characterization of Materials during 22-26 December 2014 at IIT Kanpur.

6. 8th Kudowa Summer School at Warsaw, Poland in 2009

7. DST-SERC School on "Science and Technology of Processing Plasmas" during 15-27 December 2008 at BIT Mesra, Ranchi.

RESEARCH AREAS

Plasma based surface modification processes: Plasma nitriding, Atmospheric Pressure Plasmas, DBD Plasma

AWARD AND HONOURS

1. Best paper presentation award in International Conference on Recent Trends in Engineering and Sciences (RTES-2024), organized by BIT Mesra, Ranchi during the 29-30 March 2024.

2. International Centre for Theoretical Physics (ICTP) Award for international travel support to attend Joint ICTP/IAEA Workshop at ICTP Trieste, Italy in 2009.

3. Institute of Plasma Physics and Laser Microfusion Association (IPPLM) Award for international travel support to attend 8th Kudowa Summer School at Warsaw, Poland in 2009.

PUBLICATIONS

RESEARCH PUBLICATIONS: PUBLISHED IN JOURNALS

1. Ravindra Kumar, Y. Chandra Sharma, V. V. Sagar and D. Bhardwaj, "Characterization of Low Temperature Plasma Ion Nitriding (PIN) of Inconel 600 and 601 Alloys" Iranian Journal of Materials Science & Engineering, Vol. 17, No. 2, June 2020.

2. Y. C. Sharma, Ravindra Kumar, Vajja Vidyasagar and Dheeraj Bhardwaj, "Low temperature plasma ion nitriding (PIN) of Inconel 690 alloy" Mater. Res. Ex[press 6, 026559, 2018.

3. Ravindra Kumar, Dheeraj Bhardwaj and Y. C. Sharma, "A Review on Plasma Ion Nitriding (PIN) Process" Journal on Material Science, 6(1), 31-44, 2018.

4. Ravindra Kumar, A. K. Srivastava and S. Konar, "Evaluation of Surface Properties of Various Plasma Nitrided Low Alloy Steels" Adv. Sci. Lett. Vol. 22, 3919–3923, 2016.

5. Ravindra Kumar, Ram Prakash, J. Alphonsa, Jalaj Jain, A. Pareek, P. A. Rayjada, P. M. Raole and S. Mukherjee, "Impact of forging conditions on plasma nitrided hot-forging dies and punches" Journal of Materials Science Research; Vol. 1, No. 4; 2012.

6. Ravindra Kumar, J Alphonsa, Ram Prakash, K S Boob, J Ghanshyam, P A Rayjada, P M Raole, and S Mukherjee, "Plasma nitriding of AISI 52100 ball-bearing steel and effect of heat-treatment on nitrided layer" Bull. of Material Sci., Vol.34, No.1, pp. 153-159, 2011.

PUBLISHED IN CONFERENCES

1. Attended and presented a paper title "Plasma Ion Nitriding of Non-ferrous Alloys" in International Conference on Contemporary Trends in Multidisciplinary Research and Innovation (ICCTMRI-2023), organized by BIT Mesra, Jaipur Campus during 27-28 July-2023.

2. Attended and presented a paper title "Plasma Ion Nitriding and Characterization Techniques" in International Conference on Recent Trends in Engineering and Sciences (RTES-2023), organized by SVNIT, Surat during 2-3 May-2023.

3. Ravindra Kumar, Yogesh Chandra Sharma, Vajja Vidyasagar, and Dheeraj Bhardwaj, "Wear behavior of plasma nitrided inconel 690 alloy" AIP Conference Proceedings, 2115, 030282, 2019.

4. Ravindra Kumar, Ram Prakash and S. Konar, "Plasma Nitriding System for Surface Engineering of Industrial Components" National Conference of Vacuum Electronics Device Application (VEDA)-2012, Central Electronics Engineering Research Institute (CEERI-Pilani), 21st - 24th September 2012.

5. Ravindra Kumar, Ram Prakash, J. Alphonsa, Jalaj Jain, A. Pareek, P. A. Rayjada, P. M. Raole and S. Mukherjee "Impact of forging conditions on plasma nitrided hot-forging dies and punches" in the

conference series of Recent Development in Engineering materials (RDME), which is organized by Birla Institute of Technology, Mesra Ranchi during 12th – 14th May 2011.

6. O. P. Rishi, M. Sharma, Ravindra Kumar and R. Prakash, "Case Based Reasoning for identification of Process Parameters in Plasma Nitriding Process", in the conference series of Recent Development in Engineering materials (RDME), which is organized by Birla Institute of Technology, Mesra Ranchi during 12th – 14th May 2011.

MEMBER OF PROFESSIONAL BODIES

- Life Membership of Plasma Science Society of India (PSSI).
- Life member of Association of Asia Pacific Physical Societies-Division Plasma Physics (AAPPS-DPP).

Dr. Nidhi Singh

Joined Institute on: 8-May-2006

Assistant Professor

Chemistry

PhD

CONTACT ADDRESS

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Phone Residence: 9460325070

Email: nidhis@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 14 Years

Research: 08 Years

Individual: 14 Years

PROFESSIONAL BACKGROUND

- Coordinator, Environmental Protection Committee
- Co-Mentor, Green Environment Club
- Member Cultural Committee
- Member Parakh Committee
- Member Canteen Committee

SUBJECTS OF SP-25

- Communication Skills
- Biological Science for Engineers
- Environmental Science

RESEARCH AREAS

Environmental Impact Assessment Of Wetlands, Heavy Metal Pollution, Biodiversity Conservation

AWARD AND HONOURS

(None listed in profile)

PUBLICATIONS

CONFERENCE PRESENTATIONS

1. Presented a Paper at National Conference on sustainable Chemical and Material Sciences (SCMS2016), Sponsored by UGC, DST, DRDO and BRNS, Titled "Wetland Ecology" on 5th - 6th August 2016

2. Presented a Paper at National Conference ECMSE-2012 Organised by PG Department of Environmental Science Titled "Wetland Assessment Using GIS" from 20th -22nd December 2012

3. Presented a Poster at International workshop on Chemistry for a sustainable Future Organised by Department of Chemistry University of Rajasthan (UOR) Jaipur and MNIT, Jaipur in collaboration with Green Chemistry Network Centre, Department of Chemistry University of Delhi on December 10-12 2012

4. Presented a poster at RTES-2023 Organised by SVNIT Surat held on 02 – 03 May on the Topic "Analysis of Concentration of Nitrate and Fluoride in drinking water in Kotputli and Phagi city of Jaipur district."

5. Presented a paper in ICCTMRI -2023 held at Birla Institute of Technology off Campus Jaipur (27th to 28th July, 2023) on the Topic "Impact of heavy metals on Aquatic life-A Review"

BOOK CHAPTER

1. Title: Energy consumption Trends in Jhansi

Publisher: Eco publishing house

Authors: Dr. Swati Gupta, Prof Raju Singh Khoiyangbam, Mrs Nidhi Singh

JOURNAL PUBLICATIONS

1. Journal of research in Environment and Earth Sciences

Title: Water Quality Parameters of Mansagar Lake, Jaipur Rajasthan

Vol.10-Issue 5(2024) pp:01-08

ISSN:2348-2532

2. Quest Journal

Title: Pollution in Lakes of Jaipur, Rajasthan -A Review

Volume XII Issue II

August 2024

ISSN-2319-359x

3. Educational administration; Theory and practice

Title: Heavy metal pollution and bioaccumulation in Mansagar lake

2024, 30(1) 5541-5547

DOI 53555/kuey.v3oi1.9093

PROFESSIONAL PROFILES

ORCID: https://orcid.org/0009-0006-7615-5270

LinkedIn: https://www.linkedin.com/in/nidhi-singh-55b01a51

Dr. Pooran Singh Solanki

Joined Institute on: 1-Aug-2009

Assistant Professor

Sciences

Ph.D.

CONTACT ADDRESS

Department of Bioengineering and Biotechnology, BIT Mesra, Jaipur-Campus, 27, Malviya Industrial Area, Jaipur-302017

4404, Southern Heights, UDB Apartments, Sector 22, Pratap Nagar, Jaipur

Phone Office: +91-0141-2385094 ext 324

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Email: pooransingh@bitmesra.ac.in

WORK EXPERIENCE

Teaching: 15 Years

Research: 23 Years

Individual: 23 Years

PROFESSIONAL BACKGROUND

Ph.D in Bioinformatics and Computational Biology (2011)

Scientist at BISR Jaipur (2001 – 2009)

Assistant Professor, BIT (Aug 2009 to continue)

RESEARCH AREAS

Structural Bioinformatics: Structure Based Adaptive Evolution

Genome, Transcriptome and Metagenome Analysis

Molecular based detection of pathogens

Health and Immuno-informatics

Bioinformatics Database Development

AWARD AND HONOURS

RESEARCH RECOGNITION

Honoured by Elsevier Publishing group: Appreciation certificate for my research contribution in solving the ZIKA outbreak in Rajasthan during 2021 (which comes under United Nations Global social challenges).

SESSION CHAIR

1. Chaired a Session during International Conference on Biotechnology for Sustainable Agriculture, Environment and Health (BSAEH-2021), 4-8 April 2021, Organized by MNIT, Jaipur and Biotech Research Society, India.

2. Chaired a Session during Indian Conference on Bioinformatics, Inbox-2017, organized by BISR, Jaipur, 7th to 9th November 2017.

NATIONAL ADVISORY COMMITTEE MEMBER (2018)

Member of National Advisory Committee in the National Seminar on "Bioscience: Developing the Next Generation", organized by Department of Microbiology, Sankalchand Patel University, Visngar, Gujarat, 11-12 January 2018.

JOINT ORGANIZING SECRETARY IN INTERNATIONAL CONFERENCE (2017)

INBIX'17: Organized an International Conference in Bioinformatics during 7th - 9th November 2017 as Joint organizing Secretary, entitled as "Indian Conference on Bioinformatics", (Inbox-2017). Organized by Bioinformatics Centre, BISR, Jaipur sponsored by DBT, GOI, International Supported by: International Society for Computational Biology (ISCB) and Asia Pacific Bioinformatics Network (APBioNet). Invited a Noble Laureate Prof Temple Smith, USA.

BTISnet MEET 2005

Organized an Annual Biotechnology Information System Network(BTISnet) annual meet during 2-4 Feb 2005, sponsored by DBT, Govt of India.

MERIT SCHOLARSHIP (2000)

Awarded with a merit fellowship for being among the top selected candidates in the Advanced Bioinformatics Diploma Course at Jawaharlal Nehru University by Department of Biotechnology, Govt. of India.

INVITED SPEAKERS/ TALKS

NATIONAL / INTERNATIONAL CONFERENCE SYMPOSIUM

1. Invited talk on "PERL in Bioinformatics" during a workshop on Invited talk on "Multiple Sequence Alignment", during Faculty Development Programme on "Artificial Intelligence and Machine Learning in Healthcare" jointly organized by the Department of Bioengineering and Electronics & Communication Engineering, Birla Institute of Technology Mesra, Ranchi during 27th - 31st July, 2020, sponsored by TEQIP III.

2. Invited talk on "Bioinformatics: Protein-Ligand Interactions" during International Seminar on Bioinformatics in Healthcare, 25th Jan 2020, School of life science, JNU, Jaipur.

3. Invited talk on "Alignment and Modeling Strategies", during National Conference on "Microbiology in services of Mankind" at Sankalchand Patel University, Visnagar, Gujarat; 6 Sep 2019

4. Invited talk on "Protein-Ligand Interactions", during Bioinfo-CADD-2019, BLal Institute of Biotechnology, Malviya Nagar, Jaipur, 28 Feb 2019.

5. Invited talk on "PIASP: A tool to determine structurally conserved atomic interactions in protein-ligand Complexes", Indian Conference on Bioinformatics 2019 (Inbix'19), February 22-23, 2019, HMV College, Jalandhar, India,(Invited Talk).

6. Invited Expert talk on "Bio-computing with PERL" during a workshop on "Advancements in PERL Language", sponsored by TEQIP-III, organized by RTU, Kota at Modi Institute of Technology, Kota on 6 Apr 2018.

7. Invited talk on "Bioinformatics fundamentals" during National seminar on thrust area of Biotechnology and Bioengineering, organized by Maharaja Vinayak Global University, Jaipur Institute of Biotechnology held on 22-24 Feb 2016.

8. Invited talk on "MSA and Phylogenetics" during Workshop on "Emerging Techniques in Bioinformatics and Its Application", organized by Department of Bio-Engineering, BIT Mesra, Jharkhand, India held on Nov 30 - Dec 04, 2015.

9. Invited talk "Bioinformatics" during National Workshop on Biomedical Instruments and Technologies (NWBIT-2015), organized by Department of Zoology, Centre for Advanced Studies, University of Rajasthan, Jaipur. 13-14 March 2015.

10. Invited talk on "Bioinformatics Applications", during National workshop on Applications of Computational Tools in Medicine & Biology (NWMB 2015), 24 January 2015, organized by Department of Zoology, CAS, University of Rajasthan, Jaipur on Nov 30 - Dec 04, 2015.

11. Invited talk on "Bioinformatics Trends" during the symposium on "Trends in Bioinformatics and Biotechnology" held on 2-3 August 2014 at Department of Biotechnology, ML Sukhadia University, Udaipur.

12. Invited talk on "Protein Ligand Interactions" during National Workshop on Bioinformatics entitled "In Silico Genome and Proteome Analysis" held from 8-10 March 2014 at Department of Bioscience, Banasthali University .

13. Invited talk on "Bioinformatics Analysis for Core Biologist" during the "Teacher Training Programme on Bioinformatics" held on 24 Feb 2014, organized at Modi Institute of Technology & Management, Kota and Conducted by The Society for Environment & Development, Delhi , sponsored by DBT, GOI.

14. Invited talk on "Introduction to Bioinformatics" during the seminar on "Current Trends in Bioinformatics" held on 17 October 2012 at Department of Biotechnology, ML Sukhadia University, Udaipur.

15. Invited talk on "Bioinformatics Applications" during seminar on "Bioinformatics", at Department of Bioscience College of Arts, Science & Humanities, Mody University of Science & Technology, Lakshmangarh, Sikar, Rajasthan, 27-28 Feb 2010.

16. Invited talk on "Sequence Alignment Strategies", during Workshop on "Current Trends in Bioinformatics" , CTB06, jointly organized by BIT Mesra, Kolkata Extension Centre and Computer Society of India, Kolkata Chapter, June 30 - July 1, 2006.

PARTICIPATION IN CONFERENCE/ SYMPOSIUM/ WORKSHOP

1. Participated in Workshop on "Parasite Bioinformatics" organized by Bioinformatics Centre, University of Pune, 28-31 January 2002, funded by DBT, GOI.

2. Participated in the National workshop on "Analysis and Management of Microarray Data", organized by IMTECH, Chandigarh, 27-29 October 2003, funded by DBT, GOI.

3. Participated in National Symposium on "Science and Technology for Desert Development", organized by The National Academy of Sciences, India, 2-4 December 2004, BISR, Jaipur.

4. Participated in National workshop on "Machine Learning Techniques in Functional Proteomics", organized by Bioinformatics Centre, IMTECH, Chandigarh, 18-21 Oct 2005.

5. Participated in National Symposium on Bioinformatics, organized by BIC, College of Fisheries, Mangalore, 1-2 February 2006, funded by DBT, GOI. (Paper present).

6. Participated in the Symposium on "Advances in Structural Biology & Structure Prediction" organized by CCMB Hyderabad, 11 ADNAT Convention, 23-25 February 2007.

7. Participated in the International Conference on Open Source for Computer Aided Drug Discovery (OSCADD), organized by IMTECH, Chandigarh, 22-26 March 2009.

8. Participated in 25th Annual BTISnet Meeting, sponsored by DBT, GOI, organized by National Institute of Oceanography, Goa, 27-28 Feb 2014.

9. Participated in XXVIth BTISnet Annual Conference held on 3-4 Feb 2015 at Sri Venkateshwara University, Tirupati, Andhra Pradesh, funded by DBT, GOI.

10. Participated in the workshop on "Ion Chromatography for Water & Environment" held on 15th June 2015, at BISR, Jaipur, Sponsored by Thermo Scientific.

11. Participated in the National Workshop on "PATINFORMATICS for Bioinformatics", organized by BIF, Kumaun University, Bhimtal, held on 5 March 2016.

12. Participated in XXVII BTISnet Annual Meet held on 3-4 March 2016 at BIF, Department of Biotechnology, Kumaun University, Bhimtal, funded by DBT, GOI.

13. Participated in one day workshop on "ISRO's Geo-portals for developmental planning and e-governance" organized by RRSC-W/NRSC(ISRO), CAZRI (ICAR), SRSAC(DST)-Jodhpur at BISR Jaipur on 12 April 2016.

14. Participated in 28th BTISnet Annual Meet at TATA Memorial Centre, Advanced Centre for Treatment, Research & Education in Cancer (ACTREC), Mumbai on 3-4 Feb 2017.

15. Participated in INAE Youth Conclave-2017, by Indian National Academy of Engineering (INAE), organized by BISR, Jaipur during 11-12 August 2017.

16. Participated in Symposium on "Emerging Trends in Computational Biology" organized by School of Computational and Integrative Sciences, JNU, New Delhi. 16-17 Dec 2017.

17. Participated in conference on "2018 NextGen Genomics, Biology, Bioinformatics and Technologies (NGBT), organized by SciGenome Research Foundation (SGRF), during 30 Sep - 2nd Oct 2018 at Jaipur, Rajasthan.

18. Participated in a two days workshop on "Bioprinting Technology" during 17-18 Aug 2022, conducted by Next Big Lab Innovations, Bangalore at BISR, Jaipur.

19. Participated in a workshop on "Earth Observations for Climate Services", organized by National Information System for Climate and Environment Studies(NICES), conducted at BISR, Jaipur on 12 October 2023.

20. Participated in fourth All India DIC meet at IIT Hyderabad, 2-3 May 2024.

PUBLICATIONS

LIST OF RESEARCH PAPER PUBLICATIONS IN PEER REVIEWED JOURNALS

RESEARCH PUBLICATIONS IN SCOPUS/SCI/SCIE/ESCI

1. Pawar K, Gupta P, Solanki P, Niraj R, Kothari S. Virtual Screening and Molecular Simulation of Drug Candidates Targeting the Human SLC4A4 Protein for Colorectal Cancer Therapy. Open Bioinform J, 2025; 18: e18750362408534. http://dx.doi.org/10.2174/0118750362408534251113050611

2. Pawar, K., Gupta, P.P., Solanki, P.S. et al. Downregulation of solute carrier family 4 members 4 as a biomarker for colorectal cancer. Discovery Oncology , Vol 16, Article 229 (2025). https://doi.org/10.1007/s12672-025-01948-y , Springer Nature.

3. Priyanka Sinoliya, Pooran Singh Solanki, Ravi Ranjan Kumar Niraj, Vinay Sharma, In-silico Approach to Combat Methicillin-resistant Staphylococcus aureus: Targeting RelP Protein with Inhibitor Peptide to Mitigate Drug Resistance, Current Drug Discovery Technologies; Volume 22, Issue , Year 2025, e15701638337060. DOI: 10.2174/0115701638337060250121154347.

4. Pawar, Krunal, Pramodkumar P. Gupta, Pooran Singh Solanki, Ravi Ranjan Kumar Niraj, and Shanker L. Kothari. 2025. "Targeting SLC4A4: A Novel Approach in Colorectal Cancer Drug Repurposing" Current Issues in Molecular Biology 47, no. 1: 67. https://doi.org/10.3390/cimb47010067.

5. Sinoliya, Priyanka, Solanki, Pooran Singh, Niraj , Kumar Ravi Ranjan, Sharma , Vinay, De novo Designing of the Antimicrobial Peptide as a Curative Agent for Methicillin-Resistant Staphylococcus aureus through a Computational Approach, Recent Advances in Anti-Infective Drug Discovery, volume 19, issue , pages 1-24, May 2024. ISSN: 2772-4344/2772-4352. DOI: 10.2174/0127724344297458240415113008

6. Sinoliya Priyanka, Solanki Singh Pooran, Niraj Kumar Ravi Ranjan and Sharma Vinay*, Computational Study of Antimicrobial Peptides for Promising Therapeutic Applications Against Methicillin-resistant Staphylococcus aureus, Current Computer-Aided Drug Design, volume 20, issue, pages 1-14, year Jan 2024, issn 1573-4099/1875-6697. https://dx.doi.org/10.2174/0115734099285473240101111303 (Published - 12 Jan 2024).

7. Sinoliya Priyanka, Solanki Singh Pooran, Piplani Sakshi, Kumar Niraj Ravi Ranjan*, Sharma Vinay, Anti-microbial Peptides against Methicillin-resistant Staphylococcus aureus: Promising Therapeutics, Current Protein & Peptide Science 2023; 24(2), PP [156 - 177] . DOI: https://dx.doi.org/10.2174/1389203724666221216115850 , Published - 6 Jan 2023

8. Rajpurohit, S., Solanki, P. S., Mayekar, H., Arya, H., Aradhya, R., Suravajhala, P., & Loeschcke, V. (16 Oct 2023). Tropical high-altitude insects show limited capacity to handle high temperatures. BioRxiv. September 13, 2022. (Preprint Server: BioRxiv). DOI - https://doi.org/10.1101/2022.09.10.507406

9. Chaturvedi, P., Saxena, V. L., Raina, V., Solanki, P. S., & Chaturvedi, A. (2022). Can Spices Cure Flu? A Multiple targets based Bioinformatics analysis. Research Journal of Pharmacy and Technology, 15(11), 4881-4886. DOI: https://doi.org/10.52711/0974-360X.2022.00820

10. Laha Anubhab, Prasad Anchita, Baj Kalash, Shukla Nidhi, Solanki Singh Pooran, Bandopadhyay Rajib, Nallapeta Sivaramaiah, Medicherla Mohan Krishna*, Suravajhala Prashanth*, Nigam Kumar Vinod*, Laccases as Gold Standard Enzymes for Biotechnology, Current Biotechnology, 2022; 11(3), PP 174-188. https://dx.doi.org/10.2174/2211550111666220826102014

11. Monisha Singhal; Pragya Chaturvedi; R.K. Gothwal; M.K. Mohan; P.S. Solanki (2021) Comparative study of synonymous codon usage in bacteria growing at extreme temperatures, International Journal of Bioinformatics Research and Applications, 2021 Vol.17 No.1, pp.53 - 68. ISSN: 1744-5485, (Published: 21 Mar 2021), DOI: https://doi.org/10.1504/IJBRA.2021.113966

12. Bharti Malhotra, Veenu Gupta, Pratibha Sharma, Ruchi Singh, Himanshu Sharma, Madhavi Vyas, RP Mathur, VK Mathur, Deepa Meena, Hemant Malhotra, Pragya D. Yadav, Gajanan Sapkal, Ullas PT, Gururaj Rao Deshpande, Rashmi Gunjikar, Heena Shaman, Devendra T Mourya, Nivedita Gupta, Sujit Singh, P Ravindran, Jitender Tiwari, Dimpal A. Nyayanit, Neeraj Kumar, Sameer Phalke, Anup Chugani, Sudhir Bhandari, Pooran Singh Solanki, Prashanth Suravajhala, Manila Salaria (2020). Clinico-epidemiological and genomic profile of first Zika Virus outbreak in India at Jaipur city of Rajasthan state, Journal of Infection and Public Health, 2020, Volume 13, Issue 12, Pages 1920-1926. (Dec 2020). DOI: https://doi.org/10.1016/j.jiph.2020.10.006

13. Chaturvedi,P.,Raina,V.,Solanki,P.S.,Saxena,V.L.,(2019). In silico Prediction of Anti-plasmodial Activity of Spices: Targeting Malarial Proteases, Journal of Clinical & Diagnostic Research, 13(8), KC01-KC06. (Published: Aug 2019). DOI: https://www.doi.org/10.7860/JCDR/2019/34534.13035

14. Pradeep Tiwari, Renuka Suravajhala, Sonal Gupta, Pragya Chaturvedi, Sandeep Kumar Mathur, Pooran Singh Solanki and Babita Malik (2019). A Molecular Docking and Pharmacokinetic Prediction of Thiazolidine-2, 4-dione Derivatives: Toward Novel Therapeutic Targets for Type-2 Diabetes Mellitus, International Journal for Computational Biology (IJCB), Vol.8, No.1, Oct 2019, pp. 02-08. ISSN: 2278-8115, DOI: https://www.doi.org/10.34040/IJCB.8.1.2019.01

15. Yadav, H, Gothwal, R K,Solanki, P S, Nehra, S, Sinha-roy, S, Ghosh, P (2014) Isolation and Characterization of Thermo-tolerant Phosphate-Solubilizing Bacteria From a Phosphate Mine and Their Rock Phosphate Solubilizing Abilities,Geomicrobiology Journal, vol.32 No. 6, 475-481 (2014). DOI: https://doi.org/10.1080/01490451.2014.943856

16. S. Lalwani, M. Mohan, P. Solanki, S. Singhal, S. Mathur and E. Spedicato (2013), "An Application of the ABS Algorithm for Modeling Multiple Regression on Massive Data, Predicting the Most Influencing Factors," Applied Mathematics, Vol. 4 No. 6, 2013, pp. 907-913. (2013). ISSN Print: 2152-7385 E-ISSN:1993-0445. DOI: https://doi.org/10.4236/am.2013.46126

17. P. S. Solanki, M. Krishna Mohan, P. Ghosh and S. L. Kothari. (2012) "Analysis of structurally conserved atomic interactions in structural homologs of nicotinamide adenine dinucleotide binding dehydrogenases." Journal of Computational Biology and Bioinformatics Research Vol 4, no. 3 (2012): 28-33. ISSN: 2141-2227. DOI: https://dx.doi.org/10.5897/jcbbr11.030

18. Amit N, OP Jangir, PS Solanki, Vijay S, M Krishnamohan (2008) In Silico and In Vivo Study of Lens Regeneration Under the Influence of Retinoid. J Proteomics Bioinform 1: 104-108. ISSN: 0974-276X DOI: https://doi.org/10.4172/jpb.1000016

RESEARCH POSTERS/ORAL PRESENTATIONS

1. Design of multi-epitope vaccine leads against dreadful Dengue virus using computational approach. 4th All India DIC Meet at IIT Hyderabad held on 2-3 May2024.

2. Database of Biochemical Tests of Pathogenic Enterobacteriaceae Family. BESCON-22, organized by BESI at BOSE Institute during 4-6 Nov 2022.

3. Development of a redox dye based novel and reliable method for detecting hydrocarbon utilizing bacteria directly from the soil samples. BESCON-22, organized by BESI at BOSE Institute during 4-6 Nov 2022.

4. An efficient metagenomic DNA extraction method from Thar Desert soils for exploring petroleum hydrocarbon utilizing bacterial diversity.BESCON-22, organized by BESI at BOSE Institute during 4-6 Nov 2022.

5. Database of Specific Primers and probes of water pathogens. 17th International Conference on Bioinformatics (InCoB 2018) organized by SCIS, Jawaharlal Nehru University, New Delhi, on 26-28 September 2018.

6. Activities and Progress Analysis in Bioinformatics at BTIS Sub-DIC, BISR. XXIX BTISnet Annual Meet organized at Tirunelveli, Tamil Nadu, 3-4 February,2018.

7. Utilization of Sea food waste for chitinase production by Brevibacillus Formusas BSIR-1 and Paenibacillus sp. Bisr-047. National conference of Biological Engineering Society on biological engineering in 21st century, BESCON 2017, organized by NSIT, New Delhi, September 8-9, 2017.

8. Database of Human Fungal Pathogens related to food and waterborne diseases. National Conference on Fungal Biotechnology (NCFB-2016) organized by Mycological Society of India (MSI), at BISR, Jaipur, held on 16-18 November 2016.

9. A hyper-thermostable chitinase from Paenibacillus sp. BISR-047: Its characterization and biocontrol potential.55th Annual Conference of AMI, held on 12-14 November 2014, at Tamil Nadu Agricultural University, Coimbatore.

10. Isolation and characterization of thermotolerant bacterial strains from Jhamarkotra phosphate mines producing alkaline phosphatase. 55th Annual Conference of AMI, held on 12-14 November 2014, at Tamil Nadu Agricultural University, Coimbatore.

11. Chitinase production from Paenibacillus sp. BISR-047 utilizing seafood waste as substrate under solid-state fermentation. National conference on "Bioprocess Technology" held on September 13th , 2014 at JNU, Jaipur , Rajasthan.

12. Chitinase Production and Biocontrol potential of a newly isolated chitinolytic bacterium Brevibacillus Formosus BISR-1. 54th Annual Conference of AMI held on 17-20 November 2013 at Maharshi Dayanand University, Rohtak.

13. Genotyping of antifungal compounds producing plant growth promoting rhizobacteria, Pseudomonas fluorescens. National conference on "Emerging Trend in Nanobiotechnology, held on 29-30 November 2013, at SLS, JNU, Jaipur, Rajasthan.

14. Antifungal activity of chitinolytic bacterial strain and their potential for biocontrol of important fungal phytopathogens of cumin. 53rd Annual conference of AMI, held on 22-25 November 2012 at KIIT University, Bhubaneshwar Odisha

15. Optimization of culture conditions for chitinase production by Paenibacillus sp. BISR-047. 53rd Annual conference of AMI-12, held on 22-25 November 2012 at KIIT University, Bhubaneshwar Odisha.

16. Comparative analysis of SSEs in extremophiles using CASSP. National Symposium on "Bioinformatics: Challenges in the post genomics era", sponsored by DBT, GOI, organized by BIC, University of Jammu, 2 Feb 2012.

17. Codon Bias Analysis of Extremophilic Bacteria. 51st Annual conference of AMI-2010, organized at BIT , Mesra, Jharkhand,

18. An Identification of Structurally Conserved Hydrogen-Bonding Interactions in the active site of NAD binding proteins. AIBA 2001: Conference on Biotechnology-the Science & the Business, 28-30 September 2001, IIT, Delhi

MEMBER OF PROFESSIONAL BODIES

- ISCB – International Society of Computational Biology (Member)
- APBionet – Asia Pacific Bioinformatics Network (Life Member)
- Bioclues – Bioinforatics Society (Life Member)
- APbians – Asia Pacific Bioinformatics Interaction & Networking Society (Member)
- BESI – Biological Engineering Society (Founder Member)
- MSI – Mycological Society of India (Life Member)

CURRENT SPONSORED PROJECTS

RESEARCH PROJECTS

ONGOING SPONSORED PROJECTS (EXTRAMURAL FUNDING)

1. Development of Database of Human Pathogens Related to Food and Waterborne Diseases. Sponsoring Agency: Department of Science & Technology, (Total Grant: 11.10 Lakhs)

2. Design Innovation Centre, Spoke to the Hub DIC, CCT, UOR , Jaipur, Sponsored by Department of Higher Education, Ministry of Education, Govt of India,(Total Grant: 1 Crore). Role: PI (Received 76 Lakhs).

3. Registry of Genetic Polymorphisms associated with Hormonal and Metabolic Disorders in Asian indian. Sponsoring Agency: Metabolism and Molecular Research Society, India (Total Grant: 22 Lakhs).

ONGOING IN-HOUSE PROJECTS

1. Development of Database of Biochemical Tests of Pathogenic Microbes for their rapid detection and comparative analysis.

2. NGS data analysis of Allele frequencies the genome of Drosophila melanogaster to study the thermal effect on the evolution of DNA and its adaptability.

3. Immuno-informatics Approach for Designing the Multi-epitope Vaccine Against Dengue Viruses (Under DIC sponsorship).

4. Virtual Screening and scaffold hopping of anti-inflammatory compounds.

5. Exploring the Potency of Existing Drugs against Dengue Virus Using Bioinformatics Approach.

COMPLETED SPONSORED RESEARCH PROJECTS

1. A Rapid Multiplex PCR based detection of waterborne human bacterial pathogens Agency: DST, Govt of Rajasthan, Cost: 12.12 Lakhs, Period 3 Years. (2017 - 2020).

2. Biotechnology Information System, Sub-Distributed Informatics Centre (BTIS SubDIC) Project, Sponsored By Department of Biotechnology (DBT), Govt of India. (2002 - 2007, 2007 - 2012, 2012 - 2017, 2017 - Mar 2020).

DATABASES DEVELOPED UNDER BTIS SUBDIC, DEPT. OF BIOTECHNOLOGY (DBT), GOVT. OF INDIA

1. DOMAP: Database of Aromatic and Medicinal Plants of Rajasthan

2. BISR-Primer: An Interactive Primer Designing Tool

3. DISPROLIC: Database of Interaction Specificity in Protein Ligand Complex.

4. PIASP: Protein Interaction Analyzer & Structure Parser.

5. CRAT: Codon Restriction Analytical Tool

TEXT AND REFERENCE BOOKS

BOOKS AND CHAPTERS

1. Understanding Post-translational Protein Modification through Proteomics Tools, Chapter 17, PP: 265-290, Book:Plant Proteomics, 1st Ed., Edited By: Aryadeep Roychoudhury, eBook ISBN: 9781003350453, CRC Press, (3 June 2024), DOI: https://doi.org/10.1201/b23255, (Co-authored)

2. Alleviation of salinity stress by microbes, Chap 9, PP : 145 -162, (Dec 2023) Book:Industrial Applications of Soil Microbes, Vol-2, Edited by: Shampi Jain, Ashutosh Gupta and Neeraj Verma, Publisher: Bentham Books, ISSN: 2811-0765, E-ISSN:2811-0773. DOI - https://doi.org/10.2174/9789815050264123020012. (Co-authored)

3. Mechanism of Antagonism: Hyperparasitism and Antibiosis. In: Kumar, A. (eds) Microbial Biocontrol: Sustainable Agriculture and Phytopathogen Management. Publisher:Springer, Cham. PP 257–277, 2022. Print ISBN - 978-3-030-87511-4. https://doi.org/10.1007/978-3-030-87512-1_11 . (Co-authored)

4. Biomanagement of Fusarium spp. associated with pulse crops, Editor(s): Ajay Kumar, Samir Droby, Food Security and Plant Disease Management, Elsevier Publisher, 2021, Chapter 21, Pages 423-452. ISSN 9780128218433, doi - https://doi.org/10.1016/B978-0-12-821843-3.00010-6 . (Co-authored)

5. Bio-management of Fusarium spp. associated with oil crops", Editor(s): Vijay Kumar Sharma, Maulin P.Shah, Shobhika Parmar and Ajay Kumar, Fungi Bio-Prospects in Sustainable Agriculture, Environment and Nano-Technology, Elsevier Publisher, Volume: 1: Chapter 19, Pages 475-505. ISSN: 978-0-12- 821394-0, doi- https://doi.org/10.1016/B978-0-12-821394-0.00020-2 . (Co-authored)

6. Bio-management of Fusarium spp. associated with fruit crops, Editor(s): Vijay Kumar Sharma, Maulin P. Shah, Shobhika Parmar, Ajay Kumar,Fungi Bio-Prospects in Sustainable Agriculture, Environment and Nano-Technology, Academic Press,2021,Pages 475-505, ISBN 9780128213940, https://doi.org/10.1016/B978-0-12-821394-0.00019-6 . (Co-authored)

7. Bioinformatics Tools for Microbial Diversity Analysis. In: Satyanarayana T., Johri B., Das S. (eds) Microbial Diversity in Ecosystem Sustainability and Biotechnological Applications. Springer Nature, Singapore, Vol-I, 2019, pp 23-47, DOI: https://doi.org/10.1007/978-981-13-8315-1_2 , Print ISBN: 978-981-13-8314-4, Online ISBN: 978-981-13-8315-1 . (Co-authored) Upadhyayula R.S., Solanki P.S., Suravajhala P., Medicherla K.M. (2019)

MEMBER, EDITORIAL BOARD

- BESI – Biological Engineering Society (Editorial Board Member)

MEMBER OF BOARD OF STUDIES OF UNIVERSITIES/INSTITUTION

- Member of Board of Studies in Bioinformatics, MDS University, Ajmer.
- Member of Board of Studies in Bioinformatics, Jaipur National University, Jaipur.
- Member of Board of Studies in Bioinformatics, Amity University, Rajasthan.